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Toward a green economy in China: current status and perspective in electricity generation

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Abstract

China, a leading global energy consumer, is undergoing a significant shift toward a green economy. The main factors driving this shift are ambitious objectives to achieving carbon neutrality and a growing focus on environmental sustainability. This review examines the current status of China's use of energy resources within the context of the green economy. A review of the published literature was conducted to provide updated data and information on China's energy transition from fossil fuel-based electricity generation to the utilization of renewable energy sources. The discussion concluded that although China relies mostly on fossil fuels, it has made significant progress in renewable energy penetration in the national mix. Wind and solar electricity are growing rapidly, but hydropower dominates. China has demonstrated its commitment to renewable energy by surpassing many other large countries through a rapid investment in clean energy. However, thoughtful policymaking is needed to balance economic growth with environmental protection. This work emphasizes the urgent need to prevent new coal-fired power plants, adopt management rules to permit high renewable energy generation from the national grid, and allow cheap consumer prices to drive consumption and generation. While China's efforts to become more environmentally friendly are still ongoing, it has become a global leader in sustainable energy.

Keywords: *fossil fuels, renewables, China green economy, electricity generation, energy demand*

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1. Introduction

This article describes the current situation and perspective of electricity generation in China using fossil fuels and the different renewable energy sources available in the country. China is one of the countries that is systematically increasing its energy demand, and for this reason, it will need, over the next years, the construction of new power generation plants to satisfy, in the highest possible percentage, the expected increase in electricity demand. However, for many experts, the boom in the energy demand of China is a thing of the past. While energy demand previously grew in parallel with the economy, this relationship will weaken in the future, with energy demand growing well below economic growth. One reason that explains this situation is that China has the greatest pressure to achieve high energy efficiency; another is that the country is transforming its economy from one based on heavy industry and the use of fossil fuels for electricity generation to one based on the service sector and the use of renewables for electricity production.

According to the Energy Institute of Statistical Review of World Energy 2023 report [1], most of the electricity in mainland China is produced from the consumption of fossil fuels (5,700.3TWh or 64.4% of the country's total), mainly coal (5,397.8 TWh or 61% of the country's total), which causes high pollution in large cities such as Beijing and Shanghai, among others. The electricity generated by renewables in 2022 reached 1,367 TWh, representing 14.7% of the country's total, while nuclear energy generated in 2022 reached a total of 417.8 TWh, representing 4.7% of the country's total. Because of this, the Chinese government is being pressured internationally to significantly reduce the use of fossil fuels, particularly coal, in power generation and increase the use of renewable energy to reduce the negative effects of fossil fuels on the environment and population. In China, two large hydropower plants have been connected to the grid in recent years to reduce the country's high reliance on coal for power generation. One of them is called "Three Gorges", with a capacity of 18.2 GW, and the other is called "Yellow River", with a capacity of 15.8 GW.

In 2022, electricity generation in the country was 8,848.7 TWh, representing an increase of 3.7% compared to that reported in 2021 (8,534.3 TWh) [1]. Electricity generation in China by the type of energy source in 2022 was as follows:

- **Coal**: 5,397.8 TWh (61% of the total). Compared to 2021, this represented an increase of 1.3% or 69 TWh.
- **Renewable energies**: 2,730.5 TWh (30.9% of the country's total). Compared to 2021, this represented an increase of 8.5%

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or 332.1 TWh. Regarding the weight of renewable energies within the total electricity generation at the country level, the increase was only 1.3%. According to the International Energy Agency (IEA) forecasts in the document entitled "Renewable Energy Market Update 2021 [2]", new capacities of the different types of renewable energies will continue to be built worldwide in the next two years after an unprecedented year 2020. The largest increase in newly installed renewable energy capacities was in the solar photovoltaic energy sector (48% of the total), followed by wind energy (41%) and hydropower (7%). Given its specific characteristics, wind power remains the fastest-growing source of renewable electricity generation in absolute terms. Without a doubt, wind energy is cheaper than any other renewable energy source, but care must be taken if it continues to drop in price, as this trend can be dangerous and negatively influence the expansion of this type of energy source for electricity generation in the future, according to Pastor [3] in his article entitled "Wind energy is cheaper than any other, but be careful: if it continues to drop in price, it is dangerous".

The CEO of Siemens Gamesa, Andreas Nauen, has given two important data related to the use of renewable energy for electricity generation in China. The first is that wind energy is cheaper than any other. The second and more worrying for him is that, in his opinion, wind energy should not be made too cheap because this tendency to reduce its costs would make it very difficult to make these wind platforms profitable for manufacturers, which, in turn, would reduce the possibility of investing in innovations and the construction of new wind farms over time.

- **Nuclear energy**: 417.8 TWh (4.7% of the total). Compared to 2021, this represented a decrease of 10.3 TWh. Regarding the weight of nuclear energy within the total electricity generation at the country level, the increase was 2.5%. Although the Chinese government has approved the world's largest nuclear program to increase the share of nuclear energy in the country's energy matrix, currently it remains very low.
- **Natural gas**: 290.6 TWh (3.3% of the country's total). Compared to 2021, this represented an increase of 3.5 TWh or 1.8%. In 2021, electricity generation in China using natural gas reached 287.1 TWh (3.4% of the country's total).
- **Other energy sources**: 60.4 TWh (0.7% of the country's total). Compared to 2021, this represented an increase of 10.1 TWh. In 2021, electricity generation in China through the use of other energy sources reached 50.3 TWh (0.6% of the country's total).
- **Oil**: 11.9 TWh (0.01% of the country's total). Compared to 2021, the total electricity generated in 2022 through the use of oil increased by only 0.1 TWh.

The role of fossil fuels in the country's energy mix in 2022 was very high (66.5% of the country's total), particularly the use of coal (65.1% of the country's total) for electricity generation (see **Figure 1**). Renewable energies ranked second with 30.9% of the country's total.

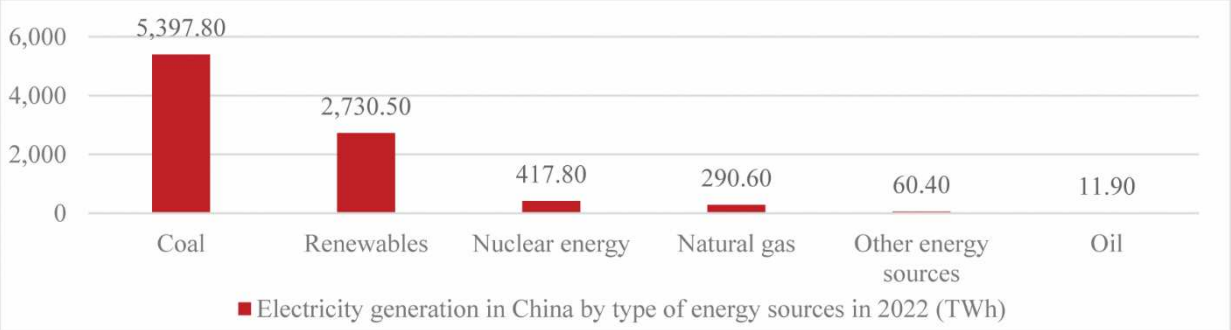


Figure 1 • Electricity generation in China by type of energy sources in 2022.
Source: Energy Institute [1].

2. Research methods

A review of the published literature was conducted to prepare this article in order to provide the reader with the most relevant updated data and information on the Chinese energy transition from a society based on burning fossil fuels for electricity generation to a society using different renewable energy sources with the same purpose.

The methodology used to prepare the manuscript is the so-called Historical-Logical Method. This method is an approach used in scientific research that combines historical analysis with logical reasoning and bridges historical context and logical reasoning, enriching the understanding of scientific phenomena. The method aims to uncover fundamental laws governing phenomena. It integrates historical data with logical reasoning, avoiding mere speculative reasoning. Like emotional intelligence, the Historical-Logical Method requires self-awareness, and researchers must objectively interpret historical events and apply logical analysis to draw meaningful conclusions.

The Historical-Logical Method enhances collaboration and encourages cohesive research cooperation, leading to better outcomes. The method recognizes that knowledge construction is dynamic and constantly evolving, allowing the author to describe the facts through their logical development.

3. Discussion

3.1. The role of fossil fuels in electricity generation in China

In this section, the analysis of the production and consumption of oil, natural gas, and coal in China is presented. How these have evolved in recent years and their expected levels for the following years, bearing in mind that fossil fuel, particularly coal, is the

most used fuel in the country for electricity generation, and the main cause of the high pollution suffered by many cities and regions in China.

3.1.1. The use of oil in electricity generation in China

According to Statista's report on proved crude oil and liquids reserves in China from 1990 to 2020 [4], oil reserves in China in 2020 were 25.1 billion barrels, almost the same as those reported since 2016. The Chinese-proven crude oil reserves represent 1.5% of the world's total proven oil reserves, ranking 14th globally.

The evolution of oil production in China during the period 2010–2022 is shown in **Figure 2**.

As illustrated in **Figure 2**, oil production increased by 0.8% throughout the period under consideration, growing from 4,077 thousand barrels per day in 2010 to 4,111 thousand barrels per day in 2022. Analyzing the data shown in **Figure 2**, it can be stated that oil production in China grew by 5.7% during the period 2010–2015, going from 4,077 thousand barrels per day in 2010 to 4,309 thousand barrels per day in 2015. From that year until 2018, oil production fell by around 12.5%, decreasing from 4,309 thousand barrels per day in 2015 to 3,798 thousand barrels per day in 2018. During the period 2018–2022, oil production in China grew again by 8.2%, increasing from 3,789 thousand barrels per day in 2018 to 4,111 thousand barrels per day in 2022.

It is important to highlight that China's oil production represented, in 2022, 4.4% of the world's total production (93,848 thousand barrels per day), which is 0.4% higher than that reached in 2021. The trend shown in **Figure 2** is expected to continue in the future. Due to the high pollution produced by burning oil for electricity generation and the low level of the country's oil reserves, it is expected that the participation of this energy source in the country's energy matrix will be lower in the future, but with a still significant weight in the transport sector.

The oil refining capacity in China in 2022 amounted to 17,259 thousand barrels per day, which represented 16.9% of the world's total, only surpassed by the refining capacity installed in the United States with 17.7% of the world's total (18,061 thousand barrels per day) [1].

The evolution of oil consumption in China during the period 2012–2022 is shown in **Figure 3**.

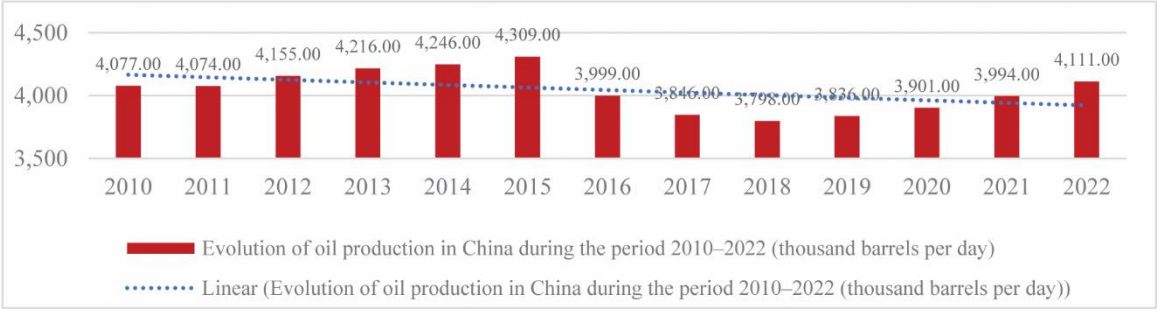


Figure 2 • The evolution of oil production in China during the period 2010–2022.
Source: BP [5] and Energy Institute [1] (for 2021 and 2022).

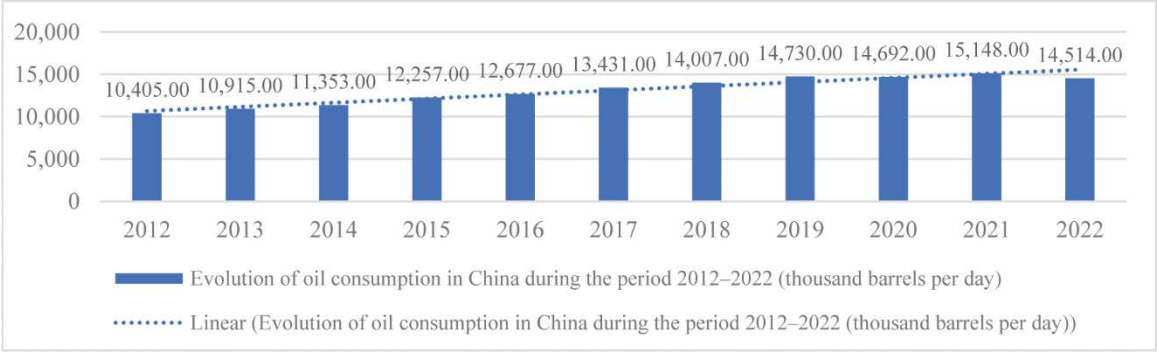


Figure 3 • The evolution of oil consumption in China, including Hong Kong, during the period 2012–2022.
Source: Energy Institute Statistical Review of World Energy 2021 and Author's calculations.

As illustrated in **Figure 3**, it can be stated that oil consumption in China in the period 2012–2022 grew by 39.5%, increasing from 10,405 thousand barrels per day in 2012 to 14,514 thousand barrels per day in 2022. This increase, together with greater consumption of coal within the country's energy matrix, had a negative impact not only on the environment but also on the population, classifying the country as the world's largest polluter of the environment. Despite the measures adopted by the Chinese government to reduce oil consumption for electricity generation, in 2022, oil consumption in the country represented 14.9% of the world's consumption of this type of fossil fuel (97,309 thousand barrels per day). This growth is estimated to continue over the next few years, although in sectors other than the electricity generation sector, such as transportation.

3.1.2. The use of natural gas in electricity generation in China

China's natural gas reserves in 2020 amounted to 8.4 trillion cubic meters, representing 4.5% of the world's total (188.1 trillion cubic meters), or 2 trillion cubic meters higher than the level reported in 2018 and the same as that reported in 2019. From 2000 to 2020, natural gas reserves in China have grown sixfold, from 1.4 trillion cubic meters in 2000 to 8.4 trillion cubic meters in 2020.

The evolution of the production and consumption of natural gas in China in the period 2010–2022 is shown in **Figures 4** and **5**.

As shown in **Figure 4**, China's natural gas production during the period 2010–2022 grew 2.3-fold, rising from 96.5 trillion cubic meters in 2010 to 221.8 trillion cubic meters in 2022. The production of this type of energy source in China in 2022 represented 5.5% of the world's total (4.043.8 trillion cubic meters). Natural gas production in China is expected to continue to grow during the coming years to reduce the use of oil and coal in power generation and high pollution in its major cities and regions.

Natural gas consumption in China during the period 2012–2022 is shown in **Figure 5**.

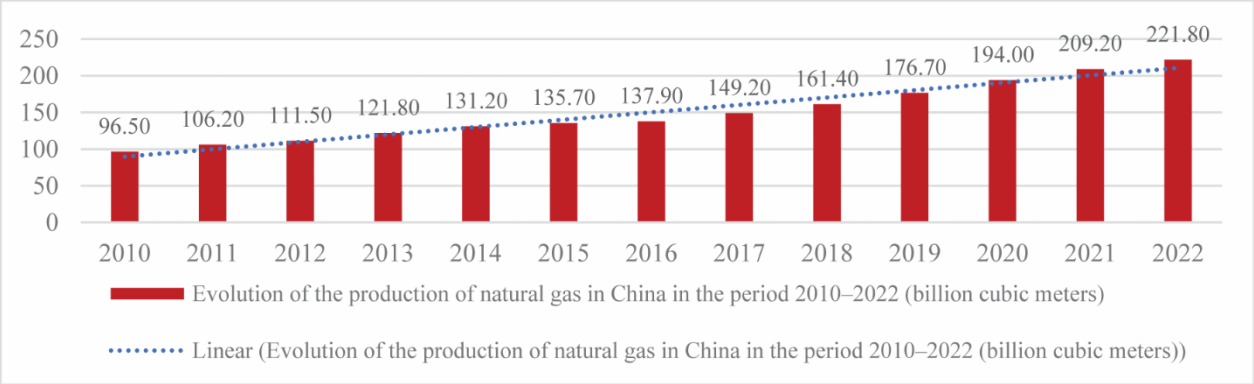


Figure 4 • The evolution of natural gas production in China in the period 2010–2022.
Source: BP [5] and Energy Institute [1] (for 2019, 2020, 2021, and 2022).

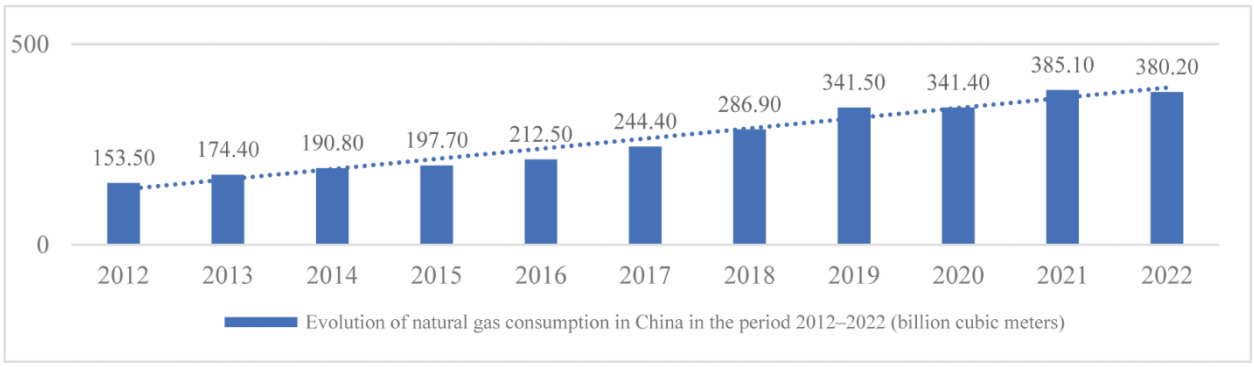


Figure 5 • The evolution of natural gas consumption in China in the period 2012–2022.
Source: Energy Institute [1] and author's calculations.

From **Figure 5**, it can be stated that the consumption of natural gas in China during the period 2012–2022 grew 2.5-fold, increasing from 153.5 billion cubic meters in 2012 to 380.2 billion cubic meters in 2022. The consumption of this type of energy source in China in 2022 represented 9.6% of the world's total (3,941.3 billion cubic meters). Natural gas consumption in China is expected to grow during the coming years to reduce the use of oil and coal in power generation and the pollution from its major cities and regions.

3.1.3. The use of coal for electricity generation in China

According to the World Population Review 2023 [6], coal reserves in China in 2023 amounted to 143,197 million tons, representing 13.3% of the world's total. China ranked fourth in world coal reserves in 2023, behind the United States with 248,941 million tons or 23.2% of the world's total, the Russian Federation with 162,166 million tons or 15.1% of the world's total, and Australia with 150,227 million tons or 14% of the world's total.

The evolution of coal production in China during the period 2010–2022 is shown in **Figure 6**.

If the whole period of 2010–2022 is considered, then it can be stated, from **Figure 6**, that coal production in China increased by 33% with three clear trends. The first was from 2010 to 2013, when coal production grew by 15.9%, increasing from 3,438.4 million tons in 2020 to 3,974.3 million tons in 2013. The second was from 2013 to 2016, when coal production decreased by 14.2%, registering the lowest production of the entire period under consideration in 2016. The third was from 2016 to 2022, when coal production grew again by 33.7%, registering in 2021 considered the highest coal production of the entire period. Regardless of these ups and downs in coal production during the 2010–2022 period, it is expected that coal production will decrease over the next few years, as well as its use in electricity generation in China in order to reduce the high levels of pollution suffered by its main cities and regions.

As shown in **Figure 7**, coal consumption in China during the period 2010–2022 grew by 20.3%, increasing from 17,562.7 million tons in 2010 to 21,130.49 million tons in 2022. The 2010–2022 period can be divided into three parts. The first part covers the period 2010–2013, when coal consumption grew by 12.6%, increasing from 17,562.10 million tons in 2010 to 19,782.50 million tons in 2013. The second part covers the period 2013–2016, with a decrease in coal consumption by 4.1%, falling from 19,782.50

million tons in 2013 to 18,969,90 million tons in 2016. The third part covers the years between 2016 and 2022, during which coal consumption in China grew again by 11.4%, increasing from 18,969.90 million tons in 2016 to 21,130.49 million tons in 2022. However, coal consumption in China is expected to decrease further during the coming years due to the closure of more obsolete and highly polluting coal-fired power plants used for electricity generation.

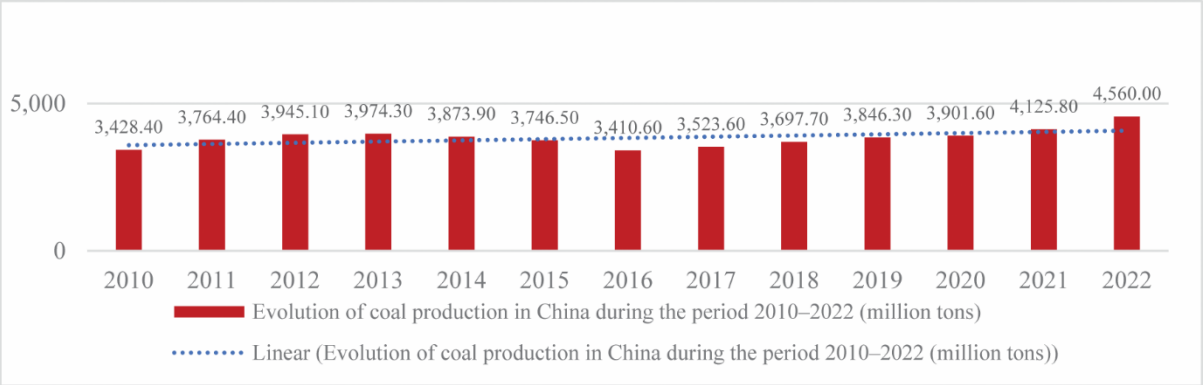


Figure 6 • The evolution of coal production in China during the period 2010–2022.
Source: BP [5] and Energy Institute [1] (for 2018, 2019, 2020, 2021, and 2022).

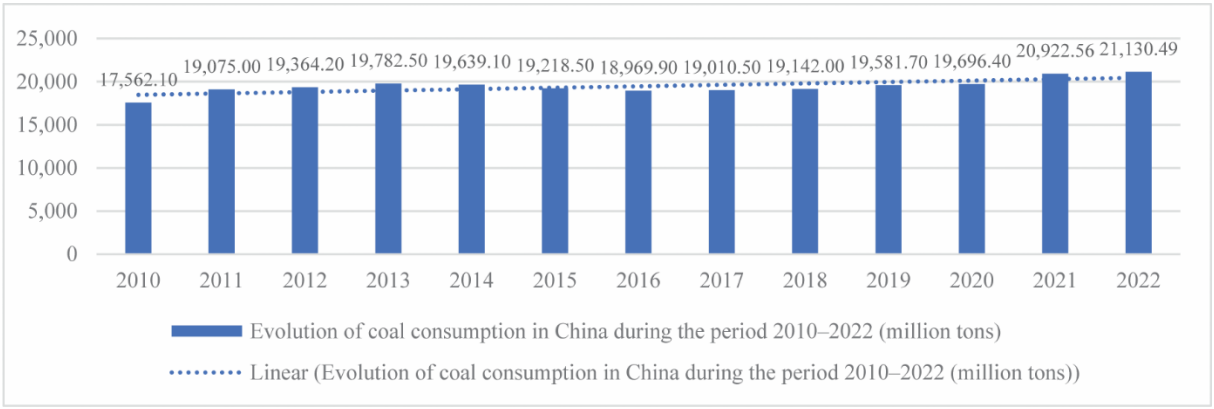


Figure 7 • The evolution of coal consumption in China during the period 2010–2022.
Source: BP [5] and Energy Institute [1] (for 2021 and 2022).

It is important to highlight that from the end of 2020, the share of coal in China's energy matrix recovered from the levels reported at the beginning of 2020. The increase in the use of coal for electricity generation in China resulted from reduced availability of electricity produced by hydroelectric power plants during the dry winter period and by rapid growth in electricity demand.

According to the document titled "Electricity Market Report July 2021 [7]", the accelerated expansion of renewable energy for electricity generation in China between July and October 2020 produced more than 210 TWh, the highest monthly level recorded in the 2015–2020 period. This expansion helped reduce coal's share of electricity generation to a new all-time low of 58% in October of that year. Unfortunately, since November 2020, the share of coal in the national energy matrix has increased again, remaining above 64% until April 2021.

During the first four months of 2021, the participation of coal within the national energy matrix was 66.5%, a higher percentage than in the same periods of 2019 and 2020. The increase in the participation of coal within the Chinese energy matrix registered as of the end of 2020 is not entirely unexpected: electricity generation through renewable energy in China is strongly conditioned by the weather, with higher electricity generation during the summer months and the rainy season in South China. Consequently, in May 2021, the greater availability of hydroelectric energy, solar photovoltaic energy, and wind energy led to a lowering of the participation of coal within the national energy matrix, falling again to 61%.

In a significant change to its energy policy, the Chinese government announced in February 2022 that it would help bring its coal-fired power plants to full capacity, raising further alarm about the fulfillment of climate pledges made by the Chinese government. Sectors of the world's second-largest economy came to a standstill in 2021 due to power shortages, partly caused by a drop in coal supplies as global fossil fuel prices soared. China, the world's biggest emitter of the greenhouse gases responsible for climate change, has pledged to peak emissions by 2030 and become carbon neutral by 2060.

Nevertheless, coal output has increased since last year's power shortages, raising uncertainty and concern about meeting those targets. According to the state news agency Xinhua, the focus on energy security and economic growth was reiterated at a high-level meeting of China's State Council, chaired by Premier Li Keqiang. At the meeting mentioned above, it was decided to increase the supply of coal and support for coal-fired power plants so that they operate at full capacity and generate more electricity to meet industrial and residential demand, thus compensating for the difficulties that have arisen with the use of

renewable energies for electricity generation due to the climatic situation of the country. The Chinese government has asked top lawmakers to ensure that emission cuts do not harm the country's economic growth and energy security, seen as a signal to limit restrictions on the use of coal for power generation for the next years.

Following the 2021 energy crisis, China reopened dozens of coal mines and removed production quotas, undoing previous steps taken to reduce greenhouse gas emissions. Due to measures taken by the Chinese government, coal production hit a record near four billion metric tons, the highest in a decade, after the pandemic disrupted imports. Two-thirds of China's economy is fueled by coal, estimated to generate 29% of the world's greenhouse gas emissions, twice the U.S. share and thrice the E.U. share.

In light of China's recent commitment to achieving zero emissions by 2060, this seasonal shift back to using coal to meet a surge in electricity demand signals the major challenge China faces, at least in the short to medium term, to achieve the reduction of electricity generation through coal-fired power plants, even when the electricity demand grows strongly, and to achieve a greater participation of renewable and nuclear energies within the national energy matrix in the coming years.

Advantages and disadvantages of using fossil fuels for electricity generation

Fossil fuels, especially oil, are the basis of the world economy and Western civilization. The massive use of fossil fuels as a source of energy began to skyrocket with the Industrial Revolution, and, in turn, the population also began to grow exponentially. Currently, most of the energy used comes from burning fossil fuels, and this activity pollutes a lot.1 It is the main responsible for climate change affecting all countries in one way or another. That is why it is necessary to consider the future of fossil energy and how to reduce greenhouse gas emissions in the atmosphere from burning fossil fuels for electricity production [8].

Fossil fuels are the main energy source used for various purposes, such as lighting, cooking, and heating. Aside from its extensive range of use, some adverse environmental effects must be considered. These adverse effects have been increasingly negative for the environment and population for several decades.

Among the possibilities available to the energy sector to reduce greenhouse gas emissions, it is worth highlighting the decrease in energy consumed and the net carbon intensity of the energy sector by changing and controlling CO_2 emissions. Fossil fuels comprise 80% of the world's current primary energy demand, and the energy system is the source of approximately 67% of global CO_2 emissions. Given that the level of emissions of methane and other short-lived climate pollutants is believed to be vastly underestimated, energy production and its use are likely the sources of an even larger proportion of emissions. If current trends continue, or if the current share of fossil fuels is maintained and energy demand nearly doubles by 2050, emissions will vastly surpass the amount of carbon that can be emitted if growth is to be limited by the global average temperature at 2°C. A high level of emissions would have disastrous climatic consequences for the planet [9].

Energy efficiency and renewable energy sources are often positioned as the only solutions available to meet climate goals in the energy system, but they are not enough. "The need to reduce emissions does not preclude the use of fossil fuels, but it does require a significant change in direction; business as usual is not consistent with decreasing emissions in global energy systems. Including an expansion of CCS will be essential, and this technology is expected to result in 16% of annual emissions reduction by 2050. This assertion is supported by the Fifth Assessment Synthesis Report of the Intergovernmental Panel on Climate Change, which estimates that limiting energy sector emissions without CCS would increase the cost of climate mitigation by 138%" [9].

All advantages and disadvantages of using fossil fuels for electricity generation must be analyzed to determine which specific type of fossil fuel should be used in any given country within the region. For this reason, using this type of energy resource will be justified in a given country if the advantages of its use are greater than the disadvantages. According to different expert opinions, the following are the main advantages and disadvantages of using fossil fuels for electricity generation.

Main advantages

- **Efficiency**: Undoubtedly, it is considered one of the most efficient energy sources for electricity production from which a significant amount of energy is produced. Therefore, a small amount of oil can provide great energy [10]. However, in the case of China, the use of oil for electricity generation in 2022 reached only 1.4% (11.9 TWh) of the total electricity produced in the country. For this reason, the use of oil for this specific purpose in China could be reduced to zero during the coming years without affecting electricity generation;
- **By-products are useful**: There is a significant number of by-products that can be obtained from fossil fuels that are very important for the economic development of the countries, such as plastics, shoes, and cleaners, among others. Plastics are widely used in several countries [10]. China is one of the countries that use oil for this specific purpose, and this situation is not expected to change in the future;
- **Transport**: Moving fossil fuel sources is easier than moving other energy sources like water, wind, and solar power. In China, oil is one of the main energy sources used in the transport sector, but this situation could change in the coming years due to the introduction of electric cars, buses, and trucks;
- **Availability**: The world has depended on conventional energy sources for generating electricity for an entire period. The efficiency adds more value to using conventional energy for electricity generation. Fossil fuels can satisfy the world population's needs and are bountiful in supply. Fossil fuels are found in almost every country, including China with important coal reserves. This aspect gives governments peace of mind, knowing they will not deplete shortly [11]. Due to the insufficient oil and gas reserves, China is one of the main countries that import

[1] China is one of the three major polluter countries at the world level.

these types of energy sources from other countries to satisfy its electricity generation needs;

- **Accessibility to reserves**: China's current access to fossil fuel reserves is still not very complicated. This means that, since it is relatively easy to extract, the economic costs of exploitation of fossil fuels are economically acceptable;
- **Its transport and storage are cheap and easy**: Unlike renewable energy, transportation and storage of fossil fuels are easy. Because fossil fuels are safe and stable, they can be transported easily and efficiently over long distances. They can be transported on trucks or ships or pumped through pipes below and above the ground. These pipes are easily laid underground and can easily transport gas or oil [11];
- **Fossil fuels are a technology that is globally developed**: There are numerous technologies available because of the presence of fossil fuels. Although this fuel source is often thought of as a way to provide transportation needs, many daily products contain items manufactured thanks to fossil fuel technologies. Even renewable fuels have a foundation built on fossil fuels [12];
- **Fossil fuels have become safer over time due to clean coal technologies**: The greenhouse gas emissions can be captured and condensed, reducing the overall amount that escapes into the atmosphere. Some technologies have come close to eliminating the production of greenhouse gases, condensing them into the water, and allowing them to be released safely into the environment [12]. Despite these new clean coal technologies, China is reducing the use of this type of energy source for electricity generation, expecting to become a green economy by 1970;
- **Employment generation:** No doubt, fossil fuel industries have created job opportunities for many people in many countries, including China. Every year, it creates new employment opportunities. So, reducing the use of fossil fuels for electricity generation will affect the world economy and lead to high unemployment levels that would affect the family and country's income source [13]. This is something that will affect China's economy as well.

Main disadvantages

- **Degrading the environment**: Using fossil fuels for electricity generation negatively affects the environment and the population. Oil and coal combustion generates harmful gases that pollute and deplete the ozone layer. Fossil fuels are responsible for causing the planet's air, land, and water pollution. Burning fossil fuels produces high carbon dioxide and causes environmental and health problems. It also leads to climate change and affects biodiversity [13]. China is one of the three main polluting countries at the world level, with a negative impact on its economy;
- **Health complication**: The burst of fossil fuels causes air pollution, leading to well-being complexities. Significant levels of air pollution cause unexpected problems in children, as it triggers asthma problems and harms the lungs. Children are the most affected since running more while playing makes them breathe more air and drink more water [13]. This is something that affects China's population significantly;
- **Fossil fuels are a finite resource**: Unlike water, wind, and solar power, fossil fuels are in a limited amount on the Earth. Coal, gas, and oil reserves will be exhausted in a given time. When that time comes, all countries should have to rely on renewable energy sources for electricity generation [11]. For this reason, China is adopting measures to change its energy mix to reduce the use of fossil fuels and to increase the use of renewables for electricity generation;
- **Cause of wars**: Fossil fuel extraction provokes wars between countries for their control [8]. Fortunately, China is not a country involved in wars for oil control;
- **Greenhouse effect on the environment and population**: The burn of fossil fuels for electricity generation, among other activities, enhanced the greenhouse effect on the environment and population; produced acid rain that harms the environment and agriculture and increases the average temperature of the planet (global warming), thus melting ice on the polar caps and rising sea levels; increase in torrential rains, the frequency of hurricanes, droughts and also floods, and increase the contamination of soil, air, surface, and groundwater. China is one of the countries more affected by global warming and, for this reason, is adopting measures to reduce this effect to the minimum;
- **Human error and accidents**: Fossil fuels can damage the environment through human error. Fossil fuels have led to extreme accidents, including pipeline leaks, exploding drilling platforms, and dumping millions of gallons of oil into the ocean. Fossil fuels will continue to pollute the planet as long as they are used. Although fossil fuels have been meeting the world's energy demands for many years, it does not mean we should continue using them as our main energy resource. Alternative energy sources are becoming cheaper and more reliable and have the added benefit of not emitting CO_2 while powering our lives [14]; China experienced 57 oil spill incidents in its coastal waters between December 29, 2018, and August 31, 2021. These spills were primarily concentrated in the Bohai Sea, the Yellow Sea, and the South China Sea;
- **Fossil fuels can be dangerous to harvest**: Coal miners can develop a condition known as "Black Lung Disease", which is almost always fatal in severe cases. Natural gas drillers can be exposed to concentrated chemicals and silica, leading to adverse health issues. Oil workers are exposed to toxic chemicals frequently, which can increase their risks of cancer development [12].

3.2. Renewable energy and power generation in China

Since 2006, China has experienced rapid growth in the use of its renewable energy resources for electricity generation, particularly wind power, placing the country among the world's leading countries in terms of installation and electricity generation through renewable energy. This growth was greatly facilitated by the renewable energy policy framework included in the Renewable Energy Law, passed by the Chinese government in 2005 and later amended in 2009 [15]. This law adopted key policies that included, among others, the following:

- Energy-specific renewable energy goal;

- A mandatory connection and purchase policy for electricity generated by renewable energy plants;
- A national system of regulated tariffs;
- Arrangements to share costs and finance renewable energy projects.

In other words, the objective of the law mentioned above was to promote the development and use of different renewable energy sources in electricity generation, increase the energy supply through the use of renewable energies, improve the country's energy structure, guarantee energy security, protect the environment, and ensure economically and socially sustainable development. The law established that the expansion of the use of renewable energies in electricity generation would be prioritized within the Chinese energy sector.

According to Zhang and Liu [16], in 2007, the State Council of China published a medium and long-term development plan for renewable energy, establishing the goal of reaching 15% of renewable energy consumption. The target of having a wind power installed capacity of 30 GW by 2020 was already achieved in 2010. For the other renewable energies, the objectives to be achieved were the following:

- **Hydroelectric power**: 300 GW. In 2022, China had an installed capacity of 413.5 GW, 113.5 GW above the approved target;
- **Wind power**: 30 GW. In 2022, China had an installed capacity of about 366 GW, 336 GW above the approved target;
- **Solar energy**: 1.8 GW. In 2022, China had an installed capacity of 393 GW, well above the approved target;
- **Bioenergy and other sources**: 30 GW. In 2022, China installed a capacity of 34.1 GW, 4.1 GW above the approved target (IRENA [17]).

According to the document mentioned above, China is the country with the largest installed capacity of power plants that work with renewable energies worldwide, reaching, in 2022, the figure of 1,160,799 MW (34.4% of the world's total[2]), followed by the United States with 351,676 MW (10.4% of the world's total), Brazil with 175,262 MW (5.2% of the world's total), India with 162,963 MW (4.8% of the world's total), Germany with 148,378 MW (4.4% of the world's total), and Canada with 105,775 MW (3.1% of the world's total) (see **Figure 8**). Off-grid, China had, in 2022, a renewable energy power plant capacity of 945,990 MW, which represented 7.6% of the world's total.

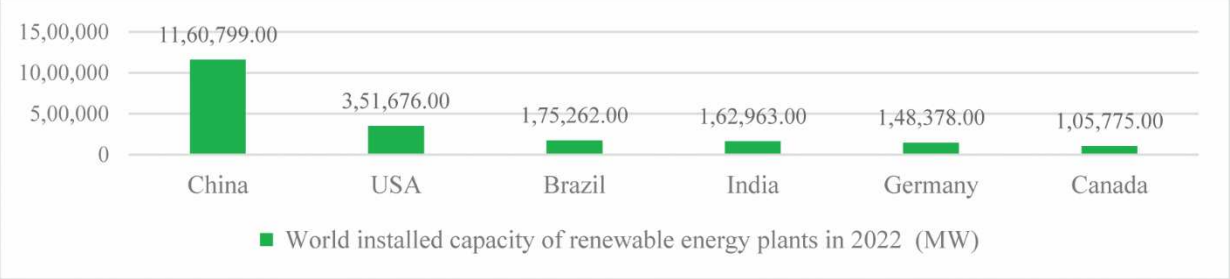


Figure 8 • Global installed capacity of renewable energy plants in 2022.
Source: IRENA [18].

In 2022, electricity generation through the use of all renewable energy sources installed in the country and reported by the Chinese competent authorities was 2,405,849 GWh or 30.6% of the total electricity produced worldwide,[3] followed by the United States with 866,274 GWh (11% of the world's total), Brazil with 507,667 GWh (6.5% of the world's total), Canada with 433,680 GWh (5.5% of the world's total), and India with 308,124 GWh (3.9% of the world's total).

According to the document "Electricity Market Report July 2021 [7]", the electricity demand in China grew by 4% year-on-year in 2020. In the first quarter of 2021, the demand for electricity in the country grew 22.5% year-on-year, in contrast to the 6% drop in the first quarter of 2020 after the blockade measures adopted by the Chinese government to contain the spread of the wave of infections caused by COVID-19. After the implementation of the strict lockdown measures by the Chinese government during the first quarter of 2020 and the subsequent drop in electricity demand, the country reached pre-pandemic demand in the second quarter of 2020, reflecting a rapid recovery of the Chinese economy. In the second half of 2020, electricity demand grew significantly beyond 2019 levels. This trend continued in the first quarter of 2021, with industrial demand higher than the first quarter of 2019 by more than 15%. The commercial demand for electricity in the first quarter of 2021 exceeded the demand registered during 2019 by 16.5% (**Figure 9**).

The projection of electricity generation through renewable energy sources in selected countries and regions during 2018–2040 is shown in **Figure 10**.

It is important to highlight that China is a world leader in electricity generation using non-traditional methods, thanks to its expansion in renewable energy use, as seen in **Figure 10**. In 2018, renewable energy in China generated 27.5% of the total energy generated in the country, being the second energy source within its energy matrix. During the period 2018–2040, China is expected to generate an annual average of 3,382 TWh, followed by India with 1,750 TWh and the European Union with 1,276 TWh.

[2]The global capacity of renewable energy plants installed and connected to the grid in 2022 reached the figure of 3,371,793 MW. Off-grid, global installed capacity amounted to 12,435,683 MW in 2022, some 3.7-fold greater than the capacity connected to the grid.

[3]The generation of electricity through the use of all renewable energy sources installed worldwide, in 2022, reached the amount of 7,857,803 GWh.

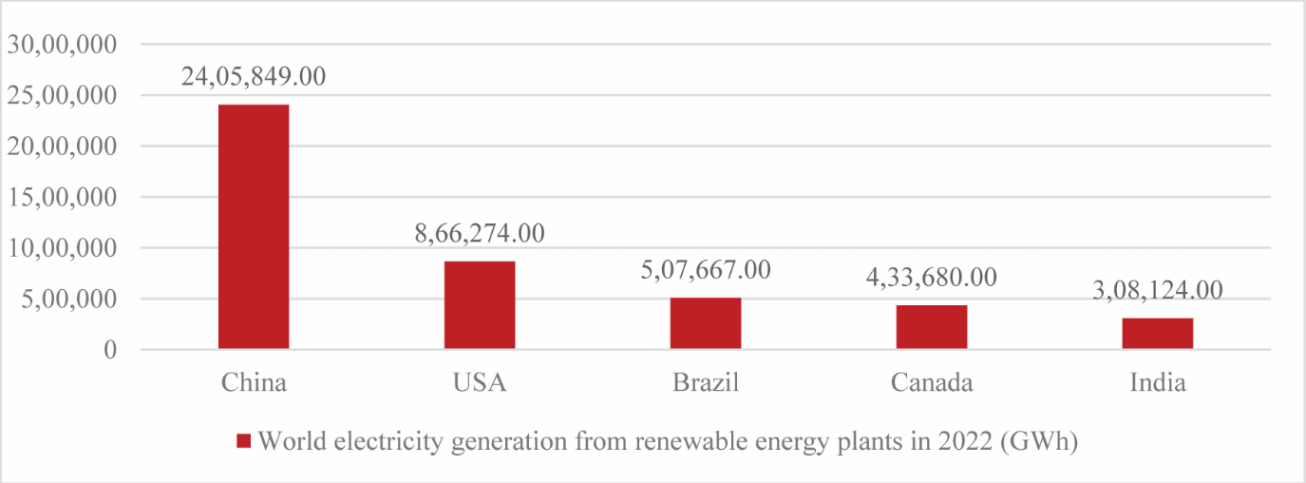


Figure 9 • World electricity generation from renewable energy plants in 2022.
Source: IRENA [18].



Figure 10 • The projection of average annual electricity generation through renewable energy in China, India, and the European Union during the period 2018–2040.
Source: World Energy Outlook 2019.

3.2.1. Hydropower in power generation in China

China is the country with the largest installed hydroelectric capacity worldwide in 2022, with 413,500 MW, representing 29.7% of the world's total (1,392,512 MW), followed by Brazil with 109,814 MW (7.9% of the world's total), the United States with 103,139 MW (7.4% of the world's total), Canada with 83,724 MW (6% of the world's total), Russia with 52,754 MW (3.8% of the world's total), and India with 52,005 MW (3.7% of the world's total) (IRENA [18]). This capacity has been growing systematically since 2010 and is expected to continue this trend during the coming years. Off-grid, China does not have hydroelectric power plants generating electricity.

The evolution of installed hydroelectric capacity in China during the period 2010–2022 is shown in **Figure 11**.

According to the data shown in **Figure 11**, it can be stated that the installed hydroelectric capacity in China during the period 2010–2022 grew by 91.4%, increasing from 216,057 MW in 2010 to 413,500 MW in 2022. The country's installed capacity growth in this type of energy source was systematic throughout the period under consideration. This has made it possible to reduce the use of coal and oil for electricity generation and thus reduce the high pollution level in many of its cities, although this reduction is still insufficient. The installed capacity of this type of energy source in the country is expected to continue growing during the coming years.

An example that confirms this trend is the following. The State Grid Corp of China, the largest state-owned utility in the country, has put into operation 2022 the world's largest pumped-storage hydroelectric plant, called "Fengning", with a capacity of 3.6 GW, consisting of 12 reversible pump-turbine units with a capacity of 300 MW each, located in the province of Hebei, some 180 km from Beijing. With a designed annual electricity generation capacity of 6,612 TWh and an annual pumped storage capacity of 8.716 billion hours, it also has the world's largest underground pumped storage plant. According to the National Energy Administration projection, the country aims to have 62 GW storage facilities operating by 2025 and 120 GW by 2030 [19].

The Fengning Pumped Storage Facility will function as a peak power plant for the safe and stable operation of the Beijing-Tianjin-North Hebei grid while balancing intermittent power supply from large wind farms and solar parks in North Hebei and Inside Mongolia. At full capacity, the Fengning Hydropower Plant is capable of storing power generated by other renewable energy sources of up to 40 million kWh, which not only helps ensure the stable and steady operation of the power grid in North China but also promotes the clean transition of the country's energy matrix, facilitating China's efforts to reach peak carbon by 2030 and carbon neutrality by 2060 [19].

The evolution of electricity production through hydroelectric plants installed in China during the period 2010–2022 is shown in **Figure 12**.

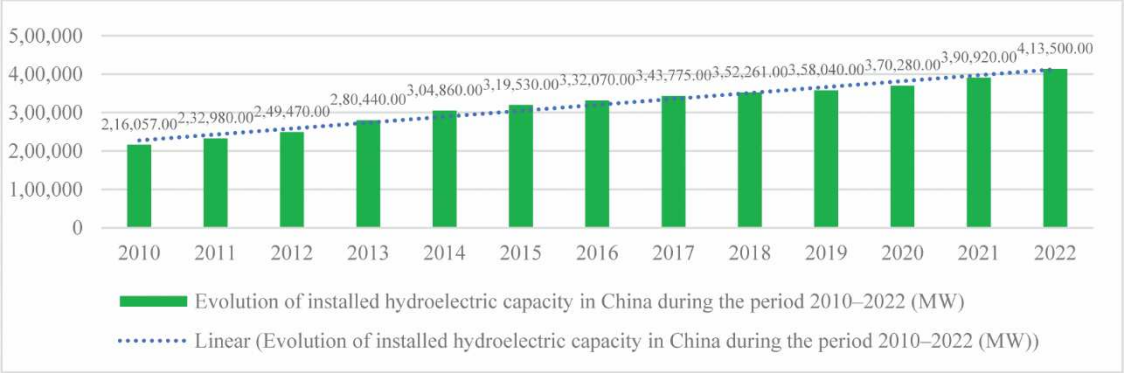


Figure 11 • The evolution of installed hydroelectric capacity in China during the period 2010–2022.
Source: IRENA [20] and IRENA [18] (for 2020, 2021, and 2022).

According to **Figure 12**, electricity generation in China through hydroelectric power plants during the period 2010–2021 grew by 85.4%, increasing from 722,170 GWh in 2010 to 1,339,000 GWh in 2021. Except for 2011 and 2021, electricity generation through hydroelectric plants grew systematically throughout the period under consideration. Electricity generation from hydropower plants in China is expected to continue growing during the coming years, as is its share of the country's energy mix as a result of the closure of a group of old and inefficient coal-fired power plants, in compliance with energy policy adopted by the Chinese government to reduce the high pollution in its main cities and regions.

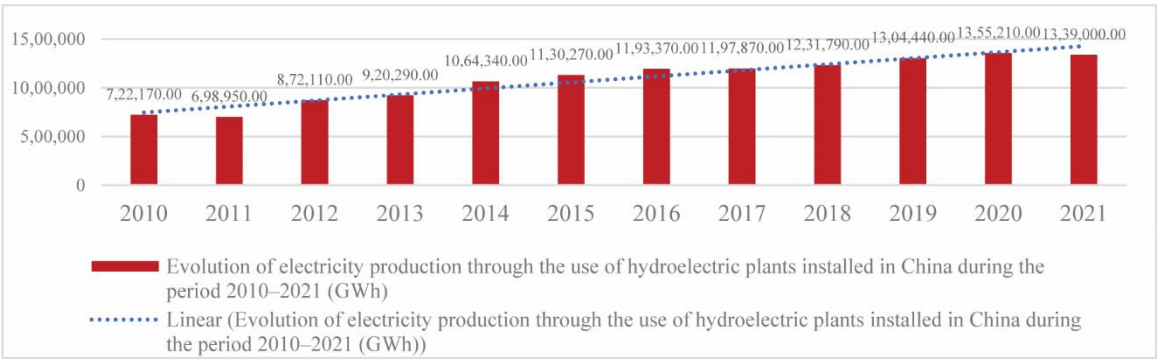


Figure 12 • The evolution of electricity production through the use of hydroelectric plants installed in China during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2020 and 2021).

However, it is important to highlight that China has the largest installed capacity of mixed hydroelectric plants worldwide in 2022, with 367,710 MW, representing 29.3% of the world's total. These plants generated, in 2021, a total of 1,300,000 GWh, or 30.4% of world electricity production through this type of facility (4,275,375 GWh). The country also has the largest installed capacity of pumped hydroelectric plants in the world, which in 2022 was 45,790 MW or 33.4% of the world's total, generating in 2021 around 39,000 GWh or 31.1% of the total worldwide (125,304 GWh). This electricity production is the highest recorded in 2021 worldwide (IRENA [18]).

Main advantages and disadvantages of using Hydroelectric energy for electricity generation

Undoubtedly, the use of hydroelectric energy for electricity generation has benefits but also consequences for the environment and the population living in the area where it will be built, so it is necessary to achieve a balance to maintain sustainability.

According to Morales Pedraza [21, 22], Maldonado [23], Pande [24], and Bhatt [25], using hydroelectric plants for electricity generation has some advantages over other energy sources. These are the following:

- **It is totally safe and clean**: Hydroelectric energy is totally ecological. Its energy production does not generate waste or release harmful substances into the environment. It is important to mention that hydroelectric energy is a good alternative to energy sources based on the combustion of gas, oil, and coal because it is friendly to nature [23].
- **Renewable energy source**: Hydropower is a renewable energy source, which means they are naturally replenished.
- **No fuel cost**: Using hydroelectric plants for electricity generation does not require any fuel to operate like most other energy sources. This is a great advantage over fossil fuels, whose costs vary at a rate that could be significant under certain conditions;
- **Low operation and maintenance costs**: Despite expensive upfront building costs, hydroelectric power is one of the cheapest energy sources. The good thing about hydroelectricity is it requires a low cost of maintenance and operation. Like other fossil-based fuel sources, such as oil or gas, there are typically no production costs linked to hydropower as water is naturally available and is free [25]. The labor cost is also usually low since hydroelectric plants

are automated and have few personnel on site during normal operation of the plants;

- **Lower cost of electricity**: The cost of electricity produced from hydropower is quite low, making it very attractive to build hydropower plants in many countries. However, an important limitation of this type of power plants is that they cannot be built in all places where electricity needs to be generated;
- **Short recovery periods of the initial investment**: In general, the recovery period of the initial investment associated with a normal-size hydroelectric plant is estimated to be between five and eight years;
- **No greenhouse gas emissions**: Electricity generation through hydroelectric plants does not produce greenhouse gas emissions or cause air pollution since they do not use fossil fuels in their operation. Because of this, hydroelectric plants are very attractive as a source of cheap electricity that does not produce carbon dioxide, although it does produce methane;
- **Energy storage**: Pumped hydroelectric storage is possible in certain types of hydroelectric plants. This makes them the ideal storage for the use of wind and solar energy for electricity generation, which are intermittent in nature and are not available at all times due to their high dependence on the weather;
- **Diversity of size**: Hydroelectric plants can be of various sizes, from micro hydroelectric plants of 1 MW capacity to large hydroelectric plants of up to 10,000 MW capacity, which makes them very versatile;
- **Efficient and Reliable**: Hydropower is one of the most efficient energy sources in the world. Solar energy is only 30–35% efficient, wind energy is 25–45% efficient, coal power is 33–40% efficient, but hydropower is up to 90% efficient. Hydroelectric power is not dependable on any outside conditions like weather conditions or oil/fuel prices; however, it depends on the amount of water flow. Hydropower is a very reliable energy source. Countries having large resources of hydropower use hydroelectricity as a baseload energy source [25];
- **Long lifespan**: Hydropower plants are considered to be a one-time investment. They have an average life span of 50–100 years, meaning one plant can support many future generations [24];
- **Hydropower resources are widely distributed throughout the world**: The potential use of hydropower plants for electricity generation exists in about 150 countries, and about 70% of the economically feasible potential remains to be developed, mainly in several developing countries [21];
- **There is no burning of fuel since it does not exist, so pollution is minimal**: Without a doubt, the electricity generated by hydroelectric plants represents the cleanest and cheapest way to produce electricity [21];
- **Support for the development of other renewable energy sources**: Hydroelectric energy is an important element of support for the development of other renewable energy sources, such as wind and solar, as it can be used together with them and serve as a base load [21];
- **Promotes energy security and price stability**: Hydroelectric plants offer stable electricity rates since electricity generation is independent of the use of any type of fuel [21];
- **Contributes to freshwater storage**: Reservoirs also offer the opportunity to improve freshwater management, enabling flood control, facilitating irrigation of the land for food production, and river transportation [21];
- **Hydropower plants are extremely flexible in their operation**: Hydropower plants allow for smooth frequency control and rapid response in emergencies [23];
- **Recreational opportunities**: The lake formed by the dam associated with a hydropower plant can be used for recreational activities like fishing, swimming, and irrigating. Dams have the capacity to store a tremendous amount of water, which can be used for various activities when there is a scarcity [24];
- **Development of rural areas:** The farther energy has to be supplied, the more difficult and costly it is. But if hydropower plants are in different places in the country, they can cover more regions. This way, the remote communities will get a huge uplift as electricity is one of the key infrastructures of development [24].

However, there are also certain disadvantages to using hydropower for electricity generation. The main disadvantages are, according to Morales Pedraza [21, 22], Maldonado [23], Robbins [26], Patrick and Chansen [27], Şentürk [28], Urban and Mitchell [29], and the World Commission on Reservoirs [30], the following:

- **Damage to the ecosystem and loss of land that could be beneficial for other purposes**: Hydroelectric plants that use dams for their operation will flood large areas of land due to the requirement for a reservoir. The large reservoirs required to operate hydroelectric plants cause the submergence of extensive areas upstream of the dams, destroying forests, swamps, and productive lowland grasslands rich in rivers and swamps. Land loss is often exacerbated by significant habitat change in surrounding areas caused by reservoir construction [26];
- **Hydropower projects can be detrimental to surrounding aquatic ecosystems, both upstream and downstream of the hydropower plant site**: Hydroelectric power generation changes the downstream river environment. The water leaving a turbine contains very little suspended sediment, which can cause scouring of riverbeds and loss of banks. As turbine gates are often opened intermittently, rapid or even daily fluctuations in river flow are observed;
- **Sedimentation and low flow**: When water flows, it can transport heavier particles downstream. This negatively affects dams and their power plants, particularly in rivers or catchment areas with high sedimentation. Sedimentation can fill a reservoir and reduce its ability to control flooding, as well as cause additional horizontal pressure on the part of the plant upstream of the dam [27, 28];
- **Changes in the amount of river flow will correlate with the amount of energy produced by a dam**: Lower river flows will reduce the amount of water stored in

a reservoir and, therefore, reduce the amount of water that can be used to generate electricity. Decreased river flow can lead to power shortages in areas that rely heavily on hydropower. The risk of flow shortages may increase as a result of climate change [29];

- **Methane emissions**: These emissions, in the case of reservoir hydroelectric plants, are due to the plant material in flooded areas decomposing in an anaerobic environment and forming methane, a greenhouse gas. Since forests were cut down in the area before the construction of the reservoir, greenhouse gas emissions from the reservoir may be higher than those from a thermal generation plant by burning oil [30];
- **Relocation of the population living in the reservoir construction area**: An important disadvantage of reservoir hydroelectric plants is the need to relocate people who live where they are planned to be built. In 2000, the World Commission on Reservoirs estimated that reservoirs built to date had physically displaced between 40 and 80 million people worldwide [30];
- **Risks of failure**: Because large conventional reservoir hydropower plants retain large volumes of water, a failure due to poor construction, natural disasters, or sabotage can be catastrophic for downstream settlements and infrastructure. Dam failures have been some of the largest man-made disasters in history. The failure of the Banqiao Dam in southern China directly caused the death of 26,000 people, and another 145,000 subsequently died from epidemics. Millions of people were left homeless. Furthermore, creating a dam in a geologically inappropriate location can cause extensive damage, such as that caused by the 1963 Vajont Dam disaster in Italy, where almost 2,500 people died;
- **High initial investment costs**: This is particularly true in the case of large hydroelectric plants [23];
- **Dependence on hydrology** (precipitation) [21];
- **Its construction is usually slow**: A hydroelectric project can take more than ten years to complete, not counting the delays that occur in construction. The execution time has countless variables that make the construction longer [23].

3.2.2. Wind power in electricity generation in China

Since 2000, electricity generation by wind farms in China has grown 21-fold. The Chinese wind market grew in 2022 by 36,699 MW of new capacities compared to what was reported in 2021 and has been the world leader in installing new wind capacities since 2008. The evolution of installed wind capacity in China during the period 2010–2022 is shown in **Figure 13**.

According to **Figure 13**, it can be stated that China has been increasing its wind capacity in recent years at a very high rate, being currently the country with the largest installed capacity worldwide with respect to this type of renewable energy.[4] The growth registered in the period considered was more than 12.3-fold, growing from 29,663 MW in 2010 to 365,964 MW in 2022, representing 40.7% of the world's total (898,856 MW).

The capacity of onshore wind farms installed in the country in 2022 was 335,504 MW or 40.6% of the world's total (836,233 MW), generating in 2021, a total of 603,994 GWh or 35.5% of the world's total (1,700,650 GWh). The capacity of offshore wind farms installed in the country in 2022 reached 30,460 MW or 48.6% of the world's total (62,623 MW), generating in 2021 a total of 52,711 GWh or 38.3% of the world's total (137,614 GWh).

The evolution of the capacity of onshore wind farms installed in China during the period 2010–2022 is shown in **Figure 14**.

According to **Figure 14**, the capacity of installed onshore wind farms in the country grew 11.4-fold during the 2010–2022 period, increasing from 29,533 MW in 2010 to 335,504 MW in 2022. China ranks first in the world regarding the capacity of its onshore wind farms, with 40.1% of the world's total.[5] It is expected that the capacity of onshore wind farms in China will continue to grow during the coming years, as well as their participation in the country's energy matrix, due to the construction of new onshore wind farms projected by the Chinese government.

In the case of offshore wind farms, China has been the world leader in constructing new offshore wind farms for the third consecutive year, with 4,070 MW of new capacities connected to the grid in 2022. The evolution of new offshore wind capacities installed in China during the period 2016–2022 is shown in **Figure 15**.

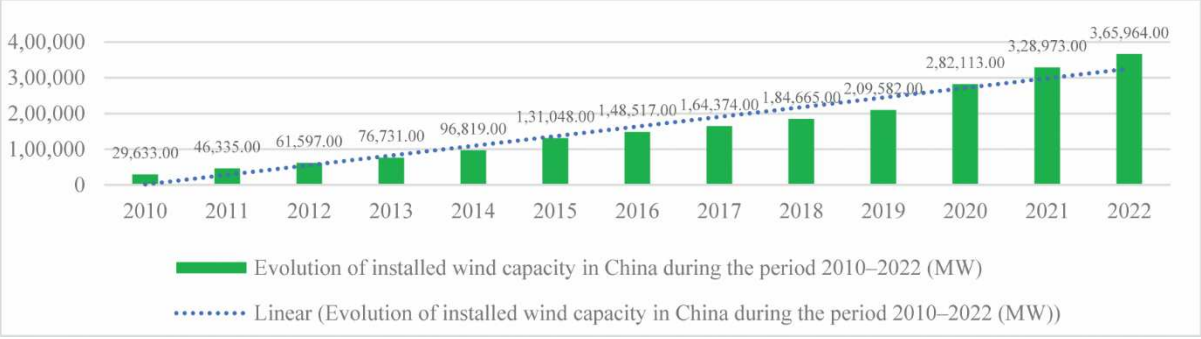


Figure 13 • Evolution of installed wind capacity in China during the period 2010–2022.
Source: IRENA [20] and IRENA [18] (for 2020, 2021, 2022).

[4] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", the global installed wind capacity in 2022 was 898,856 MW, generating a total of 1,838,264 GWh in that year.

[5] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", the total capacity of onshore wind farms installed worldwide, in 2022, amounted to 826,233 MW.

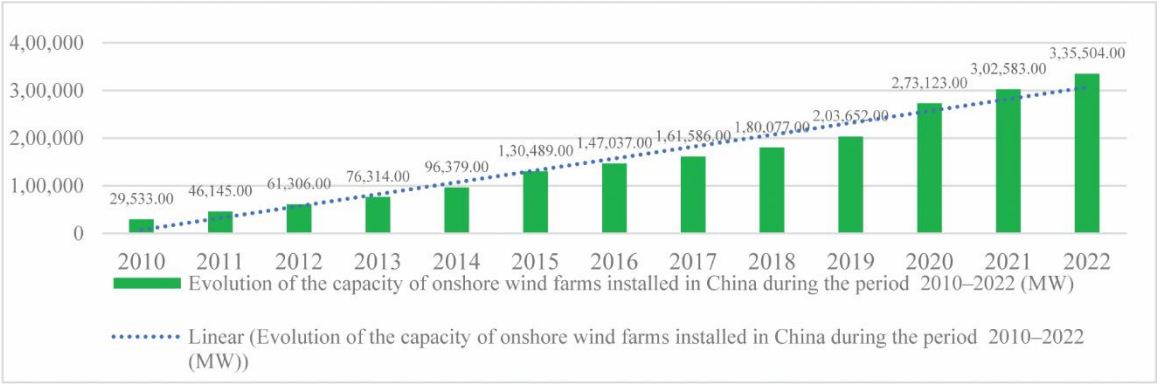


Figure 14 • The evolution of the capacity of onshore wind farms installed in China during the period 2010–2022.
Source: IRENA [20] and IRENA [18] (for 2020, 2021, and 2022).

As shown in **Figure 15**, the new offshore wind capacities installed in China in the period 2016–2022 grew by 4.6-fold, increasing from 881 MW in 2016 to 4,070 MW in 2022. The trend shown in **Figure 15** is expected to continue during the coming years. The major growth of newly installed offshore wind capacities in China was reported in 2021 with 17,400 MW.

The evolution of electricity generation reported by wind farms installed in China during the period 2010–2021 is shown in **Figure 16**.

As shown in **Figure 16**, the electricity generation from wind farms in China grew 13.4-fold during the period considered, increasing from 49,071 GWh in 2010 to 656,705 GWh in 2021. In the same year, China had the highest electricity generation through wind farms worldwide, with 35.7% of the world's total (1,834,264 GWh).

In the specific case of onshore wind farms, the evolution of electricity generation in China through this type of energy source during the period 2010–2021 is shown in **Figure 17**.

As shown in **Figure 17**, electricity generation in China from onshore wind farms grew 20.5-fold during the period considered, increasing from 29,533 GWh in 2010 to 603,994 GWh in 2021. The trend shown in **Figure 17** is expected to continue for years to come, according to plans approved by the Chinese government.

In the case of offshore wind farms installed in China, the evolution of electricity generation reported by this type of farm during the period 2010–2021 is shown in **Figure 18**.

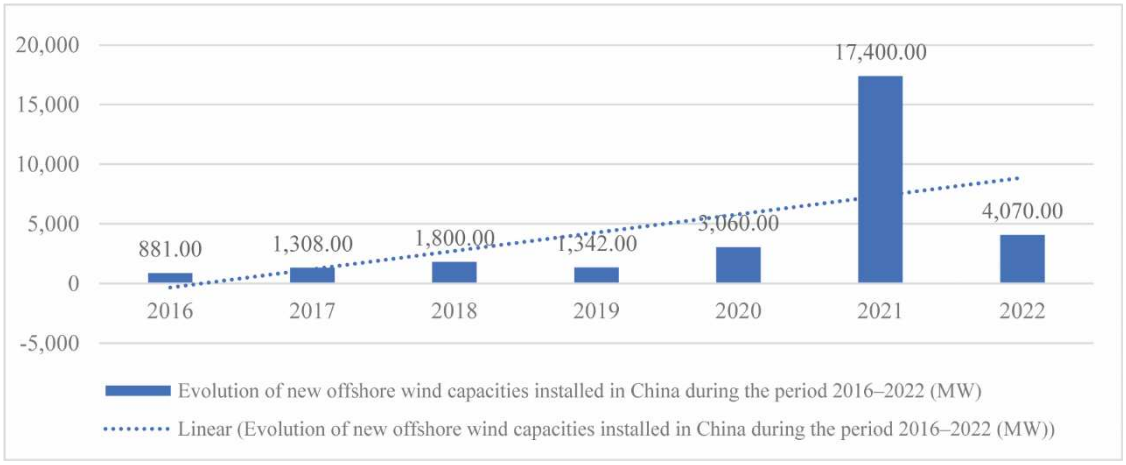


Figure 15 • The evolution of new offshore wind capacities installed in China during the period 2016–2022.
Source: IRENA ([18].

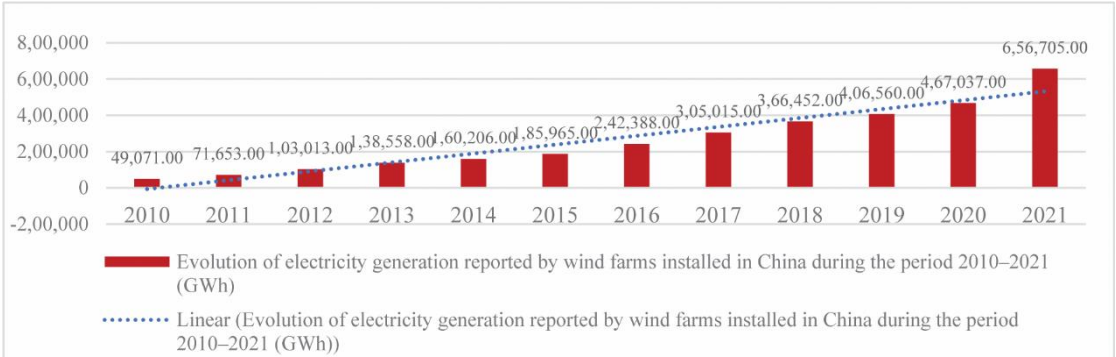


Figure 16 • The evolution of electricity generation reported by wind farms installed in China during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2019, 2020, and 2021).

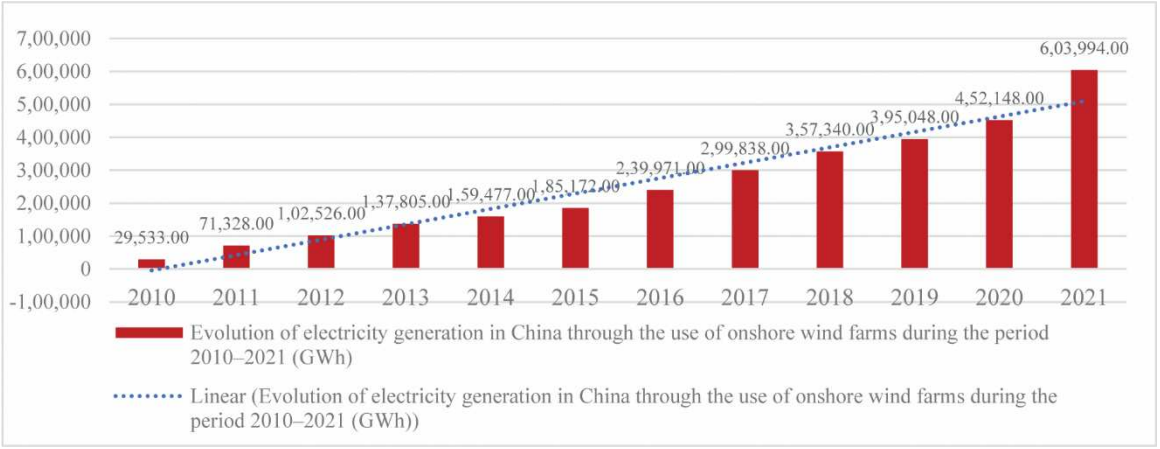


Figure 17 • The evolution of electricity generation in China through the use of this type of energy source during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2018, 2019, 2020, and 2021).

As shown in **Figure 18**, offshore wind farms installed in China during the period 2010–2021 increased electricity production by 317-fold, growing from 166 GWh in 2010 to 52,711 GWh in 2021. It is projected that the trend shown in **Figure 18** will continue during the coming years, considering the estimates of new offshore wind farm capacities that will be installed in the country during that period.

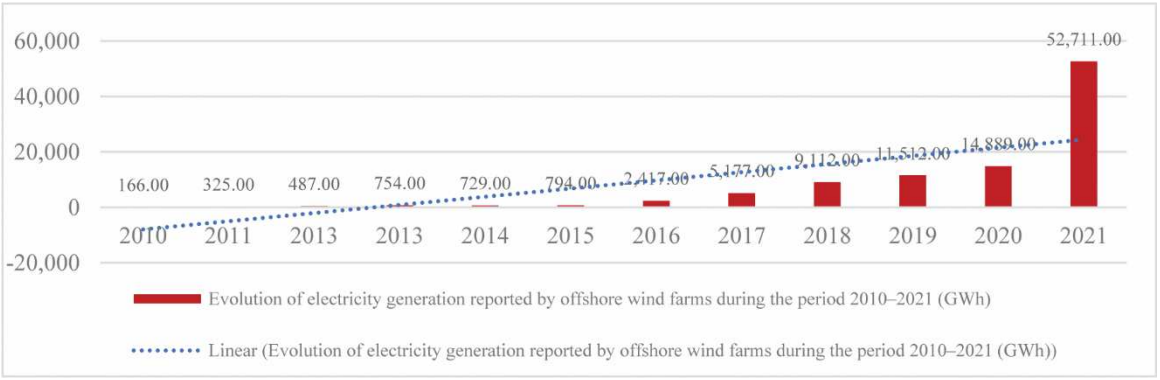


Figure 18 • The evolution of electricity generation in China through the use of offshore wind farms during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2019, 2020, and 2021).

According to the document "GWEC Market Intelligence Report 2021 [31]", the market outlook predicts that China will continue to install most of the new offshore wind capacity in the Asia-Pacific region in the first half of this decade. However, GWEC's new market forecasts indicate that China's wind market share will fall from 76% in 2021 to 56% in 2026. This decline is expected to continue until reaching a 48% share by the end of this decade.

Main advantages and disadvantages of using wind energy for electricity generation

Although it is currently estimated that approximately 3% of the electricity consumed worldwide is produced by wind farms installed in many countries. By 2040, wind energy will satisfy 9% of energy needs worldwide. However, using wind energy for electricity generation has advantages and disadvantages. These are according to Morales Pedraza [21, 22], Meg [32], and Twenergy [33], the following:

Main advantages

- **It is a clean, inexhaustible, and renewable energy source:** Wind energy is an inexhaustible and clean natural resource because you can always count on it, and in its operation, it does not generate any type of waste [22];
- **Does not require any fuel:** The turbines that generate energy are powered by the wind, so it is unnecessary to use any fuel for their operation. They are also not required to be connected to any other type of energy for them to work;
- **It is economical, and its growth is increasing**: Wind energy is cheap if compared to other energy sources used for electricity generation since wind is free, and its production costs are increasingly decreasing thanks to the technological advances that are developed to improve its efficiency, which is higher to that of solar energy. It is estimated that costs have decreased by 80% since 1980. Their participation in the energy matrix of many countries has grown by 25% annually on average [33];
- **Does not affect the environment**: The generation of electricity by wind farms does not have an aggressive impact on the composition of the soil or its erodibility. For this reason, in addition to collaborating against atmospheric pollution, wind energy plays a fundamental role in soil pollution. More and more countries require certifications to ensure that wind energy projects cause minimal side effects on the flora and fauna around the facility. In the case of wind turbines that are installed in fields, the nearby land can be used without problems for crops or for pastures that serve as food for animals;

- **It does not affect the water composition in the place where the wind farms are installed**. The operation of wind farms does not produce alterations in the water sources available in the area as they do not use water during energy production, and they do not produce waste or discharges that could contaminate the existing aquifers [33];
- **Enormous power for electricity generation**: It is estimated that if it managed to reach its full potential, the energy produced by wind farms would be 20-fold more than the electrical demand of the entire world population [33];
- **Allows supply to homes**: Wind energy, together with solar energy, can avoid connecting homes to networks for electricity supply [32];
- **It allows the use of arid and not arable land**: Wind farms can take advantage of arid lands or those not arable due to their difficult topography or deserts [32].

Main disadvantages

- **It is noisy**: Despite the technological improvements, wind turbines in wind farms still generate significant noise pollution. Therefore, it is recommended that they be built away from populated centers to avoid causing this nuisance;
- **It is an unpredictable energy source**: Because the action of the wind generates the energy, it is very difficult to plan its management since there are not always breeze currents, and you cannot even count on a certain stability in its speed. Nor can the direction be predicted in advance, as it can change suddenly;
- **It has a low effectiveness per wind turbine**: The wind density is low. For this reason, for the wind farm to be profitable, a high concentration of wind turbines operating at the same time is required;
- **There is still no efficient and economical way to store the electricity generated by a wind park**;
- **Wind farms have a negative visual impact (visual pollution)**: The construction of wind farms has an aesthetic impact on the landscape that can cause discomfort to the local population [22];
- **Wind farms put wildlife at risk**: When the wind turbines are in operation, their blades can be a real risk for birds and bats in the area since if they were to collide with them, in most cases, it would cause death [32].

3.2.3. Solar energy in electricity generation in China

China has been increasing, at a considerable rate in recent years, the installed solar capacity in the country and the generation of electricity through the use of this type of energy source. It has also increased the capacity of off-grid solar photovoltaic parks, reaching, in 2022, the amount of 421,800 MW or 9.7% of the world's total (4,340,453 MW) (IRENA [18]).

The evolution of installed solar capacity in the country during the period 2010–2022 is shown in **Figure 19**.

As shown in **Figure 19**, the capacity of the installed solar parks in China during the period under consideration grew no less than 384.6-fold, increasing from 1,022 MW in 2010 to 393.032 MW in 2022.[6] In the same year, China was the country with the largest installed solar park capacity worldwide, with 37% of the total, followed by the United States with 113,015 MW or 10.6% of the world's total, Japan with 83,055 MW or 7.8% of the world's total, and Germany with 66,664 MW or 6.3% of the world's total. It is expected, according to the plans approved by the Chinese government, to build more solar parks during the coming years to replace the production of electricity through the use of old and obsolete coal-fired power plants with much cleaner energy that reduces the high pollution of its main cities and regions, as is the case of solar energy.

In the specific case of solar photovoltaic parks, the installed capacity of this type of park in the country in 2022 reached the amount of 392,436 MW, representing 37.2% of the world's total.[7] Without a doubt, in 2022, China was the country with the largest installed capacity of solar photovoltaic parks worldwide, followed by the United States with 111,535 MW or 10.6% of the world's total, Japan with 83,055 MW or 7.9% of the world's total, and Germany with 66,662 MW or 6.3% of the world's total (IRENA [18]).

The evolution of the capacity of solar photovoltaic parks in China during the period 2010–2022 is shown in **Figure 20**.

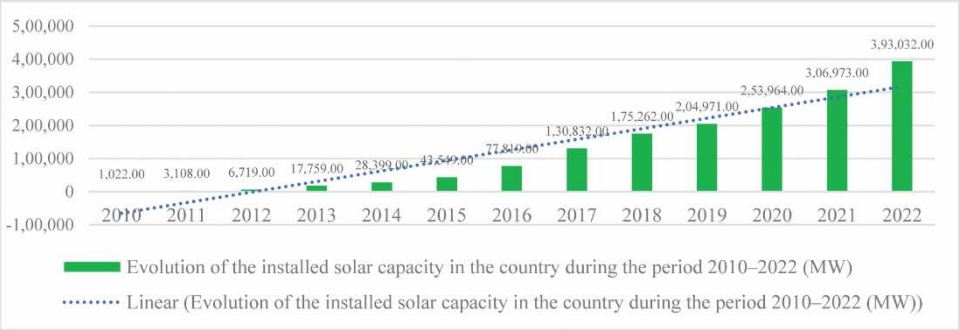


Figure 19 • The evolution of the installed solar capacity in the country during the period 2010–2022.
Source: IRENA [20] and IRENA [18] (for 2016, 2017, 2018, 2019, 2020, 2021, and 2022).

[6] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", the capacity of solar parks installed in the world, in 2022, reached the amount of 1,061,632 MW.

[7] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", the capacity of solar photovoltaic parks installed in the world amounted, in 2022, to 1,055,030 MW.

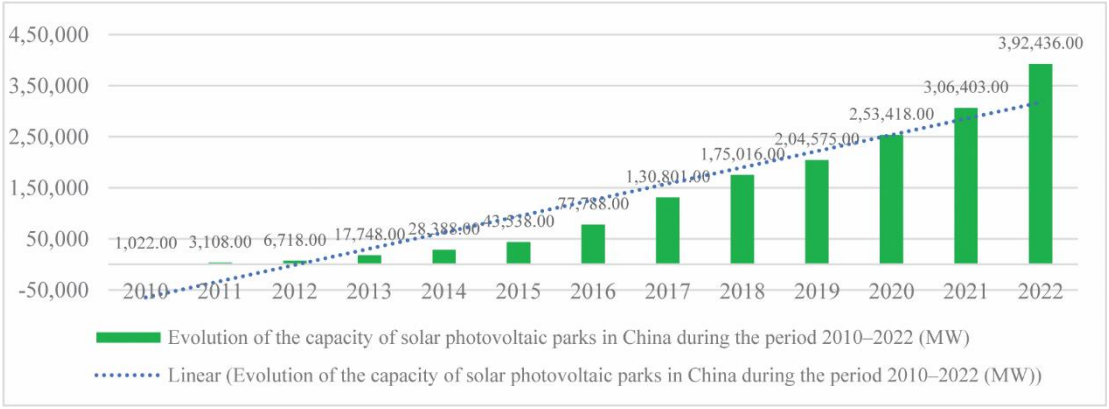


Figure 20 • The evolution of the capacity of solar photovoltaic parks in China during the period 2010–2022.
Source: IRENA [20] and IRENA [18] (for 2019, 2020, 2021, 2022).

As shown in **Figure 20**, the capacity of solar photovoltaic parks installed in China during the period 2010–2022 grew 384-fold, increasing from 1,022 MW in 2010 to 392,436 MW in 2022. It is expected that the trend shown in **Figure 20** will continue without change during the coming years in order to transform the Chinese energy matrix, where renewable energies will have to play a much greater role than the current one to reduce the high level of pollution in many of its main cities and regions.

In the specific case of solar thermal parks, the installed capacity of this type of solar park in the country in 2022 reached the amount of 596 MW, representing 9% of the world's total.[8] Undoubtedly, the installed solar thermal capacity in the country is very low compared to the solar photovoltaic parks installed in the country in the same year.

The evolution of electricity generation in China through solar parks installed in the country during the period 2010–2021 is shown in **Figure 21**.

As shown in **Figure 21**, electricity generation in China through solar parks installed in the country grew 449-fold during the period under consideration, increasing from 729 GWh in 2010 to 327,572 GWh in 2021. Electricity generation through solar parks in China is expected to continue to grow during the coming years, mainly in the case of solar photovoltaic parks, in order to continue changing the country's energy matrix, giving more weight to the use of renewable energies for electricity generation, and thus reduce the high pollution of the main Chinese cities and regions.

The evolution of electricity generation in China through solar photovoltaic and solar thermal parks during the period 2010–2021 is shown in **Figures 22** and **23**.

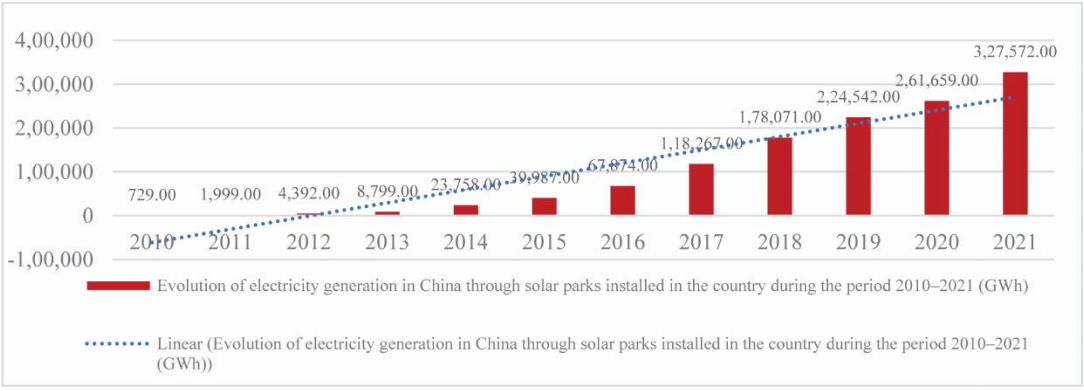


Figure 21 • The evolution of electricity generation through solar parks installed in China during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2019, 2020, and 2021).

According to **Figure 22**, electricity generation through solar photovoltaic parks installed in the country grew almost 449.3-fold, increasing from 729 GWh in 2010 to 327,551 GWh in 2021. As shown in **Figure 22**, the electricity generated by the solar photovoltaic parks installed in the country grew every year of the period considered, particularly during the last five years. The trend shown in **Figure 22** is expected to continue during the coming years, according to the energy plan already approved by the Chinese government. This plan intends to keep the country in the first place worldwide for electricity generation through solar photovoltaic parks.[9] The countries that generated most of their electricity through solar photovoltaic parks in 2021 were China with 327,551 GWh or 32.1% of the world's total, followed by the United States with 148,153 GWh or 14.5% of the world's total, Japan with 86,094 GWh or 8.4% of the world's total, and Germany with 49,340 GWh or 4.8% of the world's total.

[8] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", the capacity of solar thermal parks installed in the world amounted in 2022 to 6,602 MW.

[9] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", electricity generation through solar photovoltaic parks worldwide amounted, in 2021, to 1,020,297 GWh.

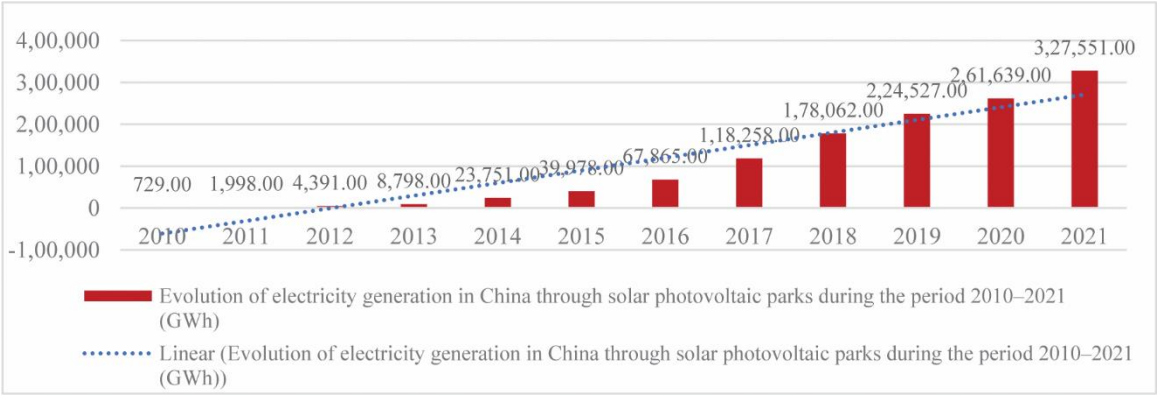


Figure 22 • The evolution of electricity generation through solar photovoltaic parks installed in China during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2020 and 2021).

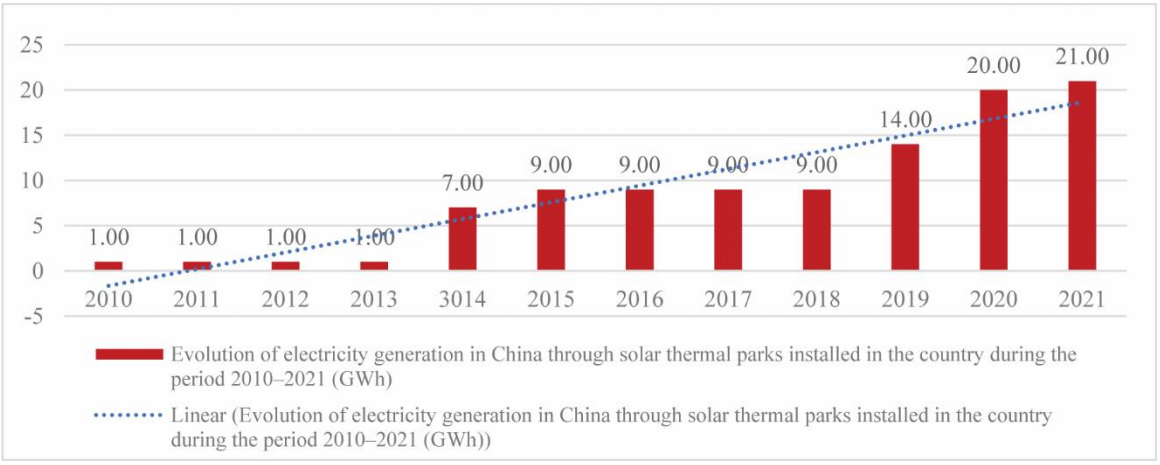


Figure 23 • The evolution of electricity generation through solar thermal parks installed in China during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2020 and 2021).

However, the evolution of electricity generation in China through solar thermal parks installed in the country during the period 2010–2021 is shown in **Figure 23**.

As shown in **Figure 23**, electricity generation in China through solar thermal parks grew 21-fold during the period considered, increasing from 1 GWh in 2010 to 21 GWh in 2021. However, the volume of electricity generated through this type of solar park in the country is very low compared with the energy generated by solar photovoltaic parks during the same period. In addition, it is important to highlight that the electricity generated by solar thermal energy within the Chinese energy matrix has remained stable from 2015 to 2018, following an increase of 2.3-fold from 2018 to 2021, as seen in **Figure 23**. No great advances are expected in electricity generation through solar thermal parks in the country during the coming years.

Main advantages and disadvantages of using solar energy for electricity generation

According to Morales Pedraza [34], Maldonado [23], and Planas (2020), the main advantages and disadvantages of using solar energy for electricity generation in China are the following:

Main advantages

- **Solar energy is a sustainable renewable energy source**: Solar energy is a resource that is not only sustainable for energy consumption. It is renewable indefinitely (at least until the Sun runs out). Solar energy can be used to generate electricity but also to heat water (solar water heaters) [34];
- **Solar panels require little maintenance**: After installation and the beginning of their operation, solar panels are very reliable because they actively create electricity in just a few millimeters and do not require any type of mechanical parts that can fail unless mobile solar parks are installed that follow the movement of the Sun, in order to increase their energy efficiency [34]. Most solar energy systems today do not require much maintenance. Residential solar panels generally only require cleaning a couple of times a year [35];
- **Solar panels are also a quiet producer of energy, a necessity when dealing with demanding neighbors**: Many governments have introduced generous tax credits for individuals and businesses that invest in solar systems for electricity generation and heating [34];
- **Solar energy is becoming less expensive every year**: With the cost of solar energy decreasing in recent years, it has become an economically competitive long-term energy source for many countries, especially developing countries;
- **Solar energy does not emit greenhouse gases that affect the environment:** Solar panels do not release greenhouse gases into the atmosphere while generating electricity. This cannot be expected from electricity generated from fossil fuels. In addition, there are efforts to find cheaper and safer technology to recycle solar panels at the

end of their useful life, which is one of the main limitations in the use of this type of energy source for electricity generation due to the high pollution of solar waste [34];

- **Solar energy is flexible in its configuration**: Solar parks for electricity generation and solar panels used to power the entire house with electricity can be mounted with different configurations and capacities on the site selected;
- **Other uses of solar panels**: Solar energy can also be used in industry, to power satellites, and even to move cars and airplanes, in order to make these means of transportation more economical and less polluting [35];
- **In remote locations, solar energy can be the ideal source of electricity**: It is cheaper to install solar panels for electricity supply in such areas than to connect them to the electrical grid [34];
- **Growing technological development:** Day by day, the solar energy industry increases its technological development, seeking to have great and fruitful improvements in the manufacture of photovoltaic cells that are more efficient, last longer, and more economical [23];
- **Use of desert regions:** Desert areas are sparsely inhabited and abandoned regions. However, they are an excellent option for installing solar parks for electricity generation [23].

Main disadvantages

- **The main disadvantage of solar energy is that it obviously cannot generate electricity at night or on very cloudy days**: The power to be generated may also be reduced during periods of cloud cover (although it is possible to produce power on a cloudy day in certain regions). A regular supply of sunlight is essential to continue generating electricity. Once the Sun sets, the level of solar energy drops very quickly. This makes it important to have an energy storage device in the form of batteries to ensure an uninterrupted electricity supply at night or on very cloudy days. If there are no batteries installed, then there will be no electricity without sunlight [34];
- **Energy production through the use of solar panels is maximized when the panel faces the Sun directly**: This means that solar panels with a fixed location will see reduced energy production when the Sun is not at an optimal angle to the solar panels. Many solar farms solve this problem by mounting the panels on mobile devices that can follow the Sun to keep the panel at optimal angles for most of the day [34];
- **Even the most efficient solar cells today only convert about 30% of most of the Sun's rays into electricity**: With increasing advancements in solar cell technology, this percentage will likely increase in the near future. In addition to their low conversion efficiency, solar panels can represent a substantial upfront investment. However, after purchase and installation, solar parks produce very cheap energy. The relationship between the amount of energy received and the amount of electrical energy obtained is low, especially compared to other energy sources such as nuclear [35];
- **Solar panels, once their useful life has ended, are sources of pollution when they are destroyed**: This element is to be considered when determining the level of solar energy participation in the country's energy matrix;
- **Requires large areas of land**: For large-capacity solar parks, the land used cannot be used for other purposes. The technology for large-scale electricity production from solar energy involves using immense areas of land that can be used for other purposes, such as agriculture or forests [23].

3.2.4. Bioenergy in power generation in China

According to the document titled "IRENA (2023) Renewable Energy Statistics 2023 [18]", China continues to lead the use of bioenergy for electricity generation within the Asia-Pacific region with a total installed capacity of 34,088 MW in 2022. This capacity represents 53.1% of the total capacity of the region (64,193 MW) and 22.6% of the world's total.[10] Other countries with the largest installed bioenergy plant capacities are Brazil with 17,206 MW or 11.4% of the world's total, the United States with 11,296 or 7.5% of the world's total, India with 10,670 MW or 7.1% of the world's total, and Germany with 9,877 MW or 6.6 % of world's total. The most important bioenergy plants installed in China consume:

- Solid biofuels and renewable waste with an installed capacity of 32,161 MW, which represents 25.4% of the world's total (126,418 MW), generating 115,514 GWh or 22.4% of the world's total (515,321 GWh);
- Renewable municipal waste with an installed capacity of 13,406 MW or 61% of the world's total (21,994 MW), generating 44,689 GWh or 51.8% of the world's total (86,204 GWh);
- Other solid biofuels with an installed capacity of 18,755 MW or 22.5% of the world's total (83,442 MW), generating 70,825 GWh or 19.1% of the world's total (370,969 GWh);
- Biogas with an installed capacity of 1,928 MW or almost 9% of the world's total (21,512 MW), generating 5,597 GWh or 6.1% of the world's total (91,688 GWh).

The evolution of the capacity of bioenergy plants installed in China during the period 2010–2022 is shown in **Figure 24**.

According to **Figure 24**, the installed capacity of bioenergy plants in China grew 9.9-fold during the period 2010–2022, increasing from 3,446 MW in 2010 to 34,088 MW in 2022. The trend in the evolution of the capacity of bioenergy plants installed in the country shown in **Figure 24** is expected to continue without change during the coming years to increase its participation in the future China's energy matrix. The countries with the largest installed capacities of bioenergy plants in Asia and the Pacific are China with 34,088 MW or 22.6% of the region's total, followed by India with 10,670 MW or 7.1% of the region's total.

[10]According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", the capacity of bioenergy plants installed worldwide reached 150,671 MW in 2022.

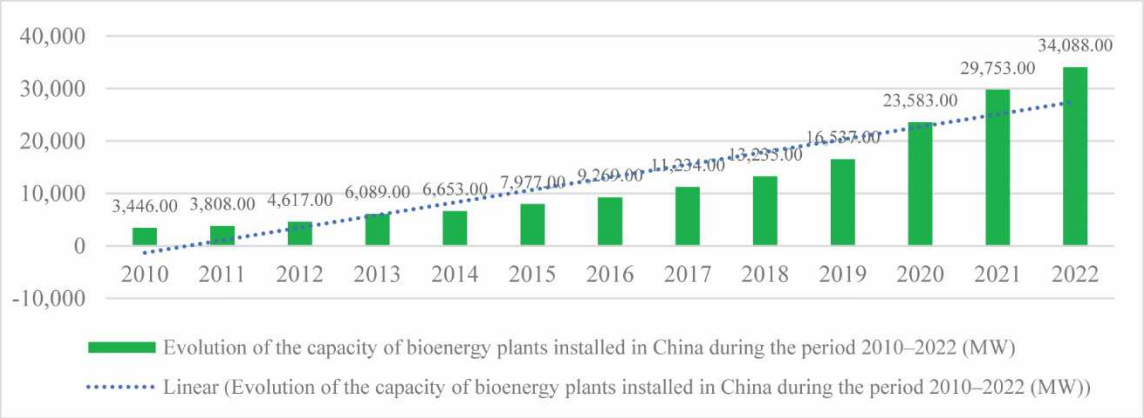


Figure 24 • The evolution of the capacity of bioenergy plants installed in China during the period 2010–2022.
Source: IRENA [20] and IRENA [18] (for 2020, 2021, and 2022).

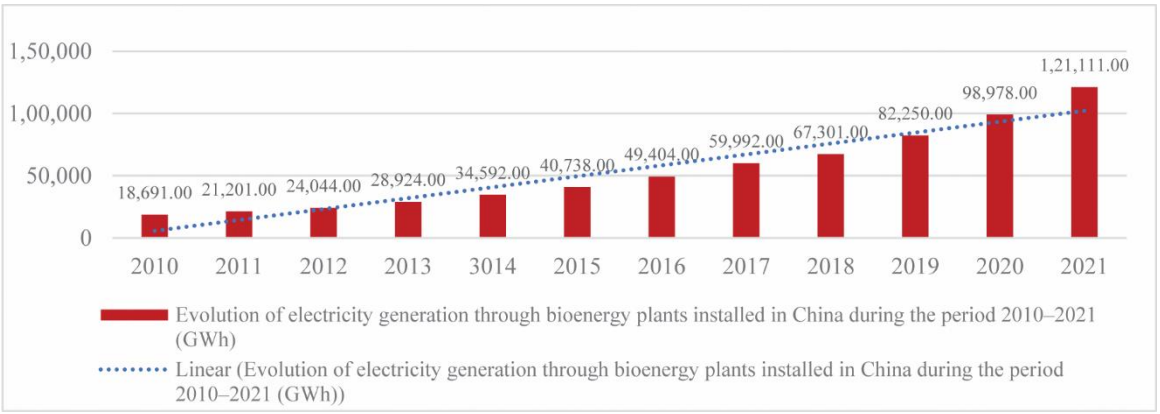


Figure 25 • The evolution of electricity generation through bioenergy plants installed in China during the period 2010–2021.
Source: IRENA [20] and IRENA [18] (for 2020 and 2021).

The evolution of electricity generation through bioenergy plants installed in China during the period 2010–2021 is shown in **Figure 25**.

As shown in **Figure 25**, electricity generation through bioenergy plants installed in the country grew 6.5-fold in the period under consideration, increasing from 18,691 GWh in 2010 to 121,111 GWh in 2021.[11] The Chinese government plans to continue building new bioenergy plants during the coming years and thus increase its participation in the country's energy matrix, aiming to reduce the high pollution in its main cities and regions.

Main advantages and disadvantages of using bioenergy for electricity generation

According to Morales Pedraza [22], Rinkesh [36], and Sawrey [37], the main advantages and disadvantages of using bioenergy for electrical production are detailed in the following paragraphs.

Main advantages

- **It is renewable and abundant**: In all countries, there is the possibility of accessing crop remains, manure, and organic waste, which in turn can be used to generate electricity;
- **Reduces emissions of CO_2 and other greenhouse gases:** Compared to thermal power plants that operate with fossil fuels, the use of biomass for electricity generation reduces emissions of CO_2 and other greenhouse gases. This is perhaps the biggest and most important advantage of generating energy from biomass. Plants capture carbon from the atmosphere during photosynthesis and become part of their structures. When the plant dies or is burned, that carbon returns to the atmosphere. Since it is a cycle, subsequent crops absorb carbon repeatedly, so a balance is maintained between the amount of carbon the biomass fuel releases into the atmosphere and the amount the plants extract from it. For this reason, fuels from biomass do not contribute to global warming and are considered clean fuels [38];
- **Carbon neutrality:** The amount of carbon released into the atmosphere is a major contributor to climate change. Biomass reduces this because the fuel is a natural part of the carbon cycle, unlike oil and other fossil fuels. The only carbon that is released into the atmosphere from biomass fuels is what was absorbed by the plants during their lifecycles [37].
- **Biomass energy is sustainable**: If crops are grown and handled in an effective way, which will be available wherever those products can be grown, then biomass is sustainable;
- **Less dependency on fossil fuels:** The more we use biomass energy, the less we need to depend on the fossil

[11] According to the document entitled "IRENA (2023) Renewable Energy Statistics 2023", electricity generation through the use of bioenergy plants worldwide, in 2021, reached the figure of 614,017 GWh.

fuels that are major contributors to climate change and other environmental issues [37].

- **Bioenergy can be used for different purposes and create various products**: This includes the production of heat, fuel for vehicles, and electricity production. Biomass energy is also versatile, as different forms of organic matter can be used to create different products. Ethanol and similar fuels can be made from corn and other crops. With so many living things on the planet, there is no limit to how many ways it can be found and used [22];
- **Biomass enables the productive use of marginal lands**: Biomass enables the use of marginal lands, such as on slopes or semi-arid lands, for productive purposes. Biomass can take waste harmful to the environment and turn it into something useful. For instance, garbage as landfill can, at least partially, be burned to create useable biomass energy [22];
- **Favors the generation of employment in rural areas**;
- **It normally requires a lower investment in foreign currency**: Compared to other energy systems, the use of the energy contained in biomass is very economical when compared to other energy sources such as fossil fuels;
- **Companies have a wide range of machinery with the most modern technology that can be used:** Exist a wide range of machinery with the most modern technology that can be used to deliver an effective resource to their clients;
- **Bioenergy helps reduce waste**: "Most waste produced in homes is either plant matter or biodegradable. This kind of waste can be channeled to more profitable use. Bioenergy generation utilizes any waste that would have otherwise found its way into landfills. Bioenergy uses this waste so that it is no longer sitting in landfills. Minimized waste means a reduction of land intended for landfills; hence, more space for human habitats" [36];
- Reduce dependency on fossil fuels: Biomass has developed as an alternate fuel source for many homeowners and has helped them reduce their dependency on fossil fuels [22].
- **Availability of trained people**: There is currently a reasonable number of trained people and institutions with knowledge of bioenergy issues.

Main disadvantages

- **It can promote extensive monoculture and reduce biodiversity**: This can affect land use for food production;
- **It can emit toxic particles during combustion with environmental implications**;
- **It can increase soil erosion and degradation**: The use of bioenergy for electricity generation can destroy habitats of great ecological value and cause the disappearance or movement of animal species by destroying their shelters and food sources;
- **Bioenergy systems generally present comparatively higher capital costs than conventional systems based on using fossil fuels for electricity generation**: This is due to the need to collect, process, and store some types of biomass;
- **Inefficiency institutional framework**: Normally, there is no efficient institutional framework to stimulate bioenergy production and rational use. This is one of the reasons for the limited use of bioenergy for electricity generation in several countries;
- **Inefficient compared to fossil fuels**: Biomass is relatively ineffective compared to fossil fuels. Ethanol, as a biodiesel, is inefficient compared to gasoline, and it often has to be mixed with some gasoline to make it work properly. On top of that, ethanol is harmful to combustion engines over long-term use [22];
- **Using biomass energy requires a lot of space**: Large amounts of land and water are needed to produce biomass crops, and when they grow, they require large spaces to be stored before they can be converted into energy [22].

Bioenergy in China has been developing rapidly, especially biomass. The installed biomass power generation capacity in China increased sharply from 6.089 MW in 2013 to 34,088 MW in 2023, representing 22.6% of the world's total, and generating in that year 121,111 GWh (or 121,299 GWh if Hong Kong is included), representing 19.8% of the world's total (IRENA [18]).

While the energy share of biomass remains relatively low compared to other renewable energy sources, China plans to increase the proportion of biomass energy by up to 15% and the total installed capacity of biomass power generation to 30 GW by 2030. Regarding impact, China's theoretical biomass energy resource is about five billion tons of coal equivalent, which equals four times all energy consumption. As per conservative estimates, currently, China is only using 5% of its total biomass potential [39].

Major biomass resources in China include waste from agriculture, forestry, industries, animal manure and sewage, and municipal solid waste. While the largest contributing sources are estimated to be residues from annual crop production like wheat straw, much of the straw and stalk are presently used for cooking and heating in rural households at low efficiencies. Therefore, agricultural residues, forestry residues, and garden waste were found to be the most cited resources with big potential for energy production in China.

Agricultural residues are derived from agriculture harvesting, such as maize, rice and cotton stalks, wheat straw, and husks, and are most available in Central and northeastern China, where most of the large stalk and straw potential is located. Because straw and stalks are produced as by-products of food production systems, they are considered sustainable sources of biomass for energy that does not threaten food security. Furthermore, it is estimated that China produces around 700 Mt of straw per year, 37% of which is corn straw, 28% rice, 20% wheat, and 15% from various other crops. Around 50% of this straw is used for fertilizers, for which 350 Mt of straw is available for energy production per year [39].

Forestry residues are mostly available in the southern and central parts of China. While a few projects that use forestry wastes like tree bark and wood processing wastes are underway, one of the most cited resources with analyzed potential is garden waste.

The Chinese government believes using biomass instead of fossil fuel for electricity generation in the residential sector could contribute to carbon neutrality as it is renewable. Although biomass consumption in China is considerable, the current status and potential of biomass use in Chinese residential households and related impacts on household biomass use have not been fully understood. As a result of a systematically reviewing literature on biomass utilization in residential homes, it is found that the current use pattern of biomass in Chinese residential homes is inefficient, not only wasting energy but also yielding amounts of air pollutants to affect climate change and human health [40].

Biogas from biomass has been encouraged for decades by the Chinese government. However, the Chinese government failed to achieve the expected goal since considerable household biogas plants were out of service. Fortunately, biogas produced from middle to large biogas plants keeps increasing, with a proportion of over 20% in China [40].

It is important to note that biomass feedstock should neither compete with edible food crops nor cause carbon debt or negative environmental impacts. As biomass plays an increasingly significant role in China's national energy mix, future research specific to technology assessment, data collection, and supply chain management of potential resources is necessary to continue understanding how biomass can become a game changer in China's energy future.

3.2.5. Geothermal energy in electricity generation in China

According to the document titled "IRENA (2023) Renewable Energy Statistics 2023 [18]", in 2022, China had a very small installed capacity of geothermal power plants, with only 26 MW reported until 2018, representing 0.2% of the world total (14,621 MW). This capacity has remained stable since 2011. Electricity generation using the geothermal energy source was also very small, reaching 144 GWh in 2022, or 0.2% of the world's total (95,251 GWh). This generation has remained stable since 2013.

However, the Chinese government has plans to more than triple the use of geothermal energy power plants for electricity generation during the coming years, and it is expected that this type of energy will represent about 1.5% of the country's total energy consumption, which will help reduce carbon dioxide emissions by 177 million tons. Geothermal resources in the country can be used for heating, electricity generation, and various other purposes.

According to the Chinese energy regulator, the geothermal power potential is equivalent to approximately 853 billion tons of standard coal and has the potential to replace coal in energy consumption. It is important to highlight that a geothermal power plant can replace 3,000 tons of coal per year, which eliminates the emission of more than 75 million tons of carbon dioxide and 30 tons of sulfur dioxide in that period.

Finally, the China Petrochemical Corporation (Sinopec), the largest geothermal energy producer in the country, pledged to continue developing this type of energy nationwide and increase the proportion of clean energy production, according to what was published by the China Daily English newspaper. The company has already replaced the use of coal with geothermal energy in ten cities in the country, according to statements by Ma Yongsheng, president of Sinopec, during the China International Geothermal Energy Forum 2018, which was held in Shanghai that year [41].

Sinopec's goal is to replace the use of coal for power generation and heating with geothermal energy in 20 cities in China and boost heating capacity to 120 million cubic meters by 2023. The regions currently using geothermal energy for heating in China have already exceeded 150 km² and, in recent years, have grown rapidly. The north of the country, which currently enjoys a sufficient supply of this type of energy, plans to have 1,000 km² of areas heated by geothermal energy.

Considering the country's abundant geothermal energy resources, representing one-sixth of the global total, very little has been exploited. Several analysts believe that the exploration and exploitation of clean energy will play a significant role in the energy optimization and security of the country during the coming years.

According to the IEA in its "Electricity Market Report July 2021 [7]" report, electricity demand grew by 4% year-on-year in China in 2020. In the first quarter of 2021, the demand grew by 22.5% year-on-year, in contrast to the 6% drop in the first quarter of 2020 following the lockdown measures imposed by countries due to the coronavirus pandemic. Compared to pre-pandemic levels in 2019, electricity demand in the first quarter of 2021 was 15% higher than in 2019, confirming the strong rebound in China's second half of 2020. With a growth of 16% in April and 15% in May 2021 compared to 2019, China continued its streak of electricity demand growth through spring 2021.

Main advantages and disadvantages of using geothermal energy for electricity generation

The main limitation for an increase in the participation of geothermal energy in the energy matrix of the countries is that this type of energy source is available only in specific places. For example, in 2019, only 31 countries were exploiting their geothermal resources. By region, geothermal energy is used for electricity generation in two African countries, six in Asia, seven in Latin America and the Caribbean, two in Eurasia, nine in Europe, two in North America, and three in Oceania.

According to Morales Pedraza [21, 22], Energía Evoluciona [42], Moriana [43], and [44], the main advantages and disadvantages of the use of geothermal energy for electricity generation are the following:

Main advantages

- **Respect the environment**: Geothermal energy is typically considered environmentally friendly. There are very few aspects related to it that can be considered polluting, but they are minimal compared to the pollution caused by the use of fossil fuels for electricity generation. The carbon footprint of a geothermal plant is minimal. Therefore, developing geothermal technology and its use for electricity generation can be considered beneficial for the fight against global warming. It is considered "clean" energy, as it reduces the consumption of fossil fuels and other non-renewable resources. It hardly produces waste, which greatly reduces the environmental impact. The greenhouse CO_2 emission is much lower than that produced by fossil fuel combustion to obtain the same energy, so it barely contributes to global warming [43]. A geothermal power plant releases, on average, the equivalent of 122 kg of CO_2 for every MWh of electricity generated, which represents

one-eighth of the carbon emissions into the atmosphere associated with conventional coal-fired power plants;

- **Renewable**: Geothermal reservoirs come from natural sources that are continuously refilled. Geothermal energy is, therefore, a form of renewable energy as long as its extraction rate is less than the natural recharge rate [43]. It can also be defined as sustainable. In other words, geothermal energy is a source that can sustain its utilization rate if properly exploited, unlike conventional sources such as coal, natural gas, or oil, whose reserves are limited. According to various studies, the energy available in the geothermal reservoirs inside the earth could last literally billions of years;
- **It is economical**: After the initial exploration stage for constructing a geothermal power plant that can be expensive, its exploitation is very economical. Its cost is economical compared to other renewable energy sources. Even after making the initial financial investment to adopt geothermal energy as an energy source, there will be significant economic and energy consumption savings. That is why, in recent years, there has been an increase in the use of geothermal energy for heating and cooling in residences and urban complexes [44];
- **Energy efficiency:** Geothermal power plants have high energy efficiency, so they can convert more geothermal heat into usable electricity compared to other technologies [42].
- **Their price is not global**: The price is set by the regions or countries where they are installed;
- **Energy performance is much higher than consumption:** In most cases, the energy production performance doubles and triples, considering the consumption value. That is, the generation system can perform more than 300% [44];
- **Constantly evolving technology:** The technology applied to geothermal energy production constantly evolves, allowing production processes to be attractive and used by everyone. Technology has even made it possible to reduce the impact of the inconveniences generated by this energy source for electricity generation and other purposes [44];
- **It is easy to control**: An advantage of geothermal energy is that it is possible to predict how much energy is going to be generated at a given time, and it is independent of the climate, which makes it very competitive concerning the use of other renewable energy sources such as solar energy or wind energy;
- **It is safe and reliable**: Geothermal power plants can operate all day without interruptions. The energy potential of geothermal can be planned and controlled, as it is a very predictable energy source. Unlike other renewable energies, geothermal energy production can be calculated very easily and with almost exact precision. This feature avoids worries about the negative consequences of atmospheric changes and many other situations [44];
- **Great potential**: Many possible geothermal deposits are not profitable, so we can only use a small part of their full potential. The most realistic estimates of production from geothermal power plants vary between 0.035 TWh and 2 TWh. Currently, geothermal power plants around the world had, in 2022, a capacity of 14,621 MW and produced about 95,251 GWh [1];
- **Stable**: Geothermal energy is a constant energy source. We can predict the production of each geothermal power plant quite accurately. This does not happen with other renewable energy sources, such as solar or wind, where climate plays a big role. Therefore, geothermal power plants can cover the base energy demand. Furthermore, these types of plants have high efficiency. The energy produced is very close to the total installed capacity, and as new technological advances continue to be achieved, better results can be expected;
- **Uses little land:** The land required for energy development through geothermal plants is much less than that used in other renewable energy sources. Additionally, it is unnecessary to carry out large-scale civil constructions with a high environmental impact [44].
- **It is ideal for heating and cooling**: To produce electricity, geothermal water sources are needed at 150° C or more so that the generating turbines installed in the plant can move. But smaller temperature gradients between the Earth's surface and interior can also be used. The Earth maintains a more or less stable temperature throughout the year. Therefore, the ground just a few meters away can function as a heat sink or source and be used like a conventional heat pump. In fact, this technology is being widely used in homes for heating/cooling.

Main disadvantages

- **Environmental aspects**: There are many greenhouse gases under the Earth's surface. Some of them can be released during the use of geothermal energy or the operation of a geothermal power plant for electricity generation and end up in the atmosphere. Geothermal power plants are associated with sulfur dioxide and silica emissions, and heavy metals such as mercury, arsenic, and boron may exist in geothermal water reservoirs. Despite all this, the atmospheric pollution associated with the operation of geothermal power plants is insignificant compared to the pollution produced by conventional thermal plants used for the same purpose;
- **Seismic risk and volcanic activity**: Some geothermal projects may increase seismic risk in areas prone to seismic or volcanic activity. That is why extreme caution must be taken when drilling wells and injecting water into the subsoil. Careless activity could trigger earthquakes or volcanic eruptions in some cases [44];
- **Environmental impacts**: Like any installation, the construction of geothermal power plants and associated infrastructure can have environmental impacts, such as the destruction of natural habitats, the alteration of the landscape, and the disturbance of local ecosystems (in short, being invasive). They impact the landscape since the construction of facilities to extract heat from underground rocks and magma requires modifications to the terrain [43]. Additionally, the disposal of geothermal fluids used in the generation process can pose challenges to properly managing and preventing water pollution [44]. Energy generation through geothermal energy can produce carbon

dioxide and sulfuric acid emissions. Both gases, harmful to the environment and humans, are generated in small values compared to other energy sources. It is important to note that sulfuric acid has a very characteristic odor (similar to rotten egg) that is easy to perceive; however, when it is generated in large quantities, this stench is not detectable, and its inhalation is lethal [44];

- **Contaminates nearby aquifers:** The operation of geothermal power plants can contaminate nearby aquifers;
- **It is expensive**: Geothermal projects are expensive. New deposits must be explored and drilled; even if the stated objective is not achieved, half of the approved budget must be paid. For each geothermal power plant with a capacity of 1 MW, an investment of between 2 and 7 million euros must be foreseen. Heat pumps for domestic use are more affordable, although they involve a large outlay at the beginning, estimated between 3,000 and 10,000 euros, but with the savings they represent in homes, the investment can be recovered after 10–20 years, depending on each case;
- **Limited geographic availability**: Geothermal energy is viable only in areas with accessible, high-temperature geothermal fields. Not all regions have access to this energy source, which limits its use to very specific places [42];
- **The transmission of electricity generated by geothermal power plants and its storage is complex:** Due to their location, which is generally far from consumption centers, the transmission of electricity generated by geothermal power plants and its storage is a complex issue;
- **Requires very detailed terrain studies and the use of specialized machinery:** The use of geothermal energy should be encouraged after carrying out details studies of the site using specialized machinery;
- **It is not transportable**: Geothermal energy is not transportable and must be used on-site, that is, in the same place where it is produced (local supply) [43].
- **Sustainability aspects**: Rainwater filters through the Earth's surface to geothermal deposits can last thousands of years. Certain studies show that reservoirs can be depleted if water is extracted at a faster rate than it is regenerated. For this reason, in some geothermal power plants, the water is reinjected once its geothermal energy has been used and electricity has been generated.

4. Conclusion

The current and future structure of the global energy matrix is not especially beneficial for combating climate change and reducing emissions of polluting gases into the atmosphere. In 2022 alone, 39,315 million tons of carbon dioxide have been emitted into the atmosphere, 7.4% more than in 2012, when the objective pursued was its reduction. China has contributed 61% to this increase, and although its progress in controlling pollution in its cities and regions is still notable, it is still insufficient. It is important to bear in mind that the very dynamics of its high economic growth in recent decades and the weight of its heavy industry have prevented this increase from being much lower.

The closure of old and inefficient coal-fired power plants, the increased use of renewable energy in electricity generation, and the bigger use of nuclear energy for the same purpose have helped reduce pollution in major Chinese cities and regions, but the measures adopted have not been enough. Other emerging economies, especially India, have also failed to make significant progress in reducing pollution from their major cities and regions, making it difficult to meet the goals set by the international community to combat climate change and reduce the current high pollution from countries and regions. "This need to reconcile the legitimate desire for growth of emerging economies with the control of environmental pollution will mark the global economy in the next decade" [45].

Coal consumption in Asia and the Pacific reached 130.50 exajoules in 2022 and 161.57 exajoules worldwide. In the same year, coal consumption in the Asia and the Pacific region represented 81% of the world's total. In the case of China, coal represented, in 2022, 61% of the total energy consumption. In 2021, the role of coal in the Chinese energy matrix was 62.4%. The reduction in the weight of this energy source within the country's energy matrix is not enough, considering the high pollution of the main Chinese cities and regions. To further reduce the role of coal within the country's energy matrix, the government has already begun decarbonizing its economy. As a result of the measures adopted, coal has gone from 68.1% of total energy consumption in 2010 to 60.2% in 2018 [45] and 61% in 2022, according to the Energy Institute [1].

In other words, there has been a decrease of 7.1% in the use of this highly polluting energy source for electricity generation and other purposes in China. However, as a result of the recent measures taken by the Chinese government to fully utilize the capacities of the country's existing coal-fired power plants for electricity generation during the coming years, it is expected that the reduction in participation of coal in the Chinese energy matrix will be greater than in the past with the implications that this would have for the decarbonization process of the Chinese energy matrix mentioned above scheduled for 2060.

An important factor for the conformation of the future Chinese energy matrix is the so-called "energy intensity". The expected evolution of energy intensity globally will depend critically on what happens in China. This country already plays a key role if we consider that between 2010 and 2018, it has contributed by 28.5% and 60.9% to the global increase in energy consumption and greenhouse gas emissions, respectively. Without a doubt, China will continue to be a key player in global energy intensity forecasts "since it is expected to contribute 30% of energy savings between 2018 and 2030, more than the sum of the United States and Western Europe (16.7% and 7.4%, respectively). It should be noted that China plans to focus its energy savings on a notable reduction in energy intensity of around 20% (higher than the 17.4% corresponding to the period 2010–2018), thanks to a process of structural transformation toward an economic model with a greater weight of the tertiary sector" [45].

China has ambitious plans to increase the use of renewable and nuclear energy for power generation over the coming decades in order to reduce, as much as possible, the use of coal in the power sector.

In the specific case of solar energy, China's goal is to reach a solar thermal installed base of at least 27.6 GW by 2030, making it the world leader in using this type of renewable energy for electricity generation. According to Abengoa [46], the strategic program "CSP Demonstration Program" was launched by the Chinese

government and articulated in successive waves of projects, the first of which was made up of 20 projects, for a total of 1.35 GW. Its initial objective was to have them executed by the end of 2018. Due to different factors, only five of these initial projects have been executed with an installed capacity of 300 MW. The two main causes of these delays are the difficulties of the developers awarded the projects to reach the necessary financing agreements within the Chinese market and the slowness in incorporating partners that contribute technology and know-how. Another of the program's key objectives was reducing the cost per kWh, which would come from competition between Chinese component suppliers. Considering the expected cost reduction, for the second wave of projects that started in 2019, about 60 projects were identified for a total of 14.9 GW, which are in the feasibility study phase, intending to reach a cumulative 5 GW for the period 2020–2021.

The Chinese program to increase the use of solar energy in the country's energy matrix foresees the installation of new capacities, mainly in solar photovoltaic parks. Only in 2017, solar photovoltaic parks with a capacity of 53 GW were installed, more than half of the world's capacity, expanding the already installed capacity by 69% (130.5 GW). The country aims to generate 20% of its energy from renewable sources by 2030. Recently, China activated, in the city of Huainan in the province of Anhui, the largest floating solar plant in the world [47], with a capacity of 40 MW. Another plant of this type, but with a capacity of 20 MW, also came into operation in the province mentioned above, generating 850 MWh to meet the electricity needs of 200,000 homes.

According to the National Energy Administration of China, the nation's cumulative installed PV capacity reached 540 GW at the end of October 2023. In the first ten months of that year, the country added 142.5 GW of new PV systems, with 13.62 GW deployed in October alone. New solar power installations hit 78.42 GW in China at the end of June 2023 [48].

Without a doubt, China is advancing by leaps and bounds in the use of future technologies and increasing the use of solar and wind energy in power generation like no other country in the world. With its growth policy in the renewable energy sector, China has now clearly replaced Germany and Europe as the world leader in using this type of energy for electricity generation.

China can achieve its dual goals of peak emissions before 2030 and carbon neutrality before 2060, accelerating while strengthening its energy security. China already leads the world in energy transition investment, accounting for US$550 billion in spending in 2022, or around half of the global total. To stay on track for net zero by 2060, China must triple the investment rate in the latter half of this decade to an annual average of US$1.66 trillion [49].

China is the world's largest wind power market in new and cumulative installations. In 2022, the country installed 36,991 MW of new wind power capacities, of which 32,921 MW are onshore wind farms and 4,070 MW are offshore wind farms. The continuous growth of this market for wind turbines is mainly due to ambitious policies at the national level that have made wind power and other renewable energies extremely cost-competitive. On May 21, 2019, the National Development and Reform Commission of China (NDRC) released a new policy that presents a clear roadmap toward greater use of subsidy-free onshore wind power. "From 2020 to 2030, 24 GW of wind turbines will be needed annually, 19 GW land-based and 5 GW offshore. Over the same period, a total of 39 GW of wind power units need to be retired or transformed. Between 2030 and 2050, average annual wind turbine needs will be about 50 GW, including 44 GW land-based and 6 GW offshore. About 400 GW of wind turbines will need to be retired or reconstructed over the period 2030–2050" [50].

According to the China National Energy Administration's new policy release, more than 60 onshore wind power projects have already been approved but have not yet been installed. After 2020, new onshore wind power facilities in China are expected to have a total capacity of 20 GW [31].

As of mid-2019, China had more than 10 GW of offshore wind farm projects under construction, with another 40 GW already approved and ready to build. Like onshore wind power, offshore wind power installation will increase in China during the coming years. Looking further into the future, the Chinese offshore wind market is expected to maintain the growth momentum recorded in recent years, with annual new capacity installations exceeding 4 GW.

China was the world's largest offshore wind farm market for new installations in 2017 and ranked second after the United Kingdom. The situation changed in 2018. That year, China installed 1.8 GW, displacing the United Kingdom (1.3 GW) for the first time as the world's largest offshore wind farm market.

In 2019, Chinese offshore wind capacities accounted for 21.1% of the global total, below the United Kingdom with 34.8% and Germany with 26.7%. In 2020, Chinese offshore wind capacities accounted for 26.2% of the world's total, surpassing Germany's offshore wind capacities. The United Kingdom continued to be the country with the largest offshore wind capacity worldwide, with 30.2% in 2020. However, in 2022, China had the world's largest installed onshore and offshore wind capacities worldwide, surpassing the United Kingdom for the first time in the wind market.

However, several studies reveal that solar and wind power are the cheapest energy sources in most parts of the world, but the race to achieve cheaper wind turbines, for example, is having a dangerous effect on the future of this technology. The low prices associated with wind energy and the growing competitiveness in this market have reduced profit margins significantly. There are also risks stemming from the worldwide shortage of raw materials due to the crisis caused by the coronavirus pandemic, which has made it difficult for suppliers to meet demand and prices for materials such as steel, widely used in these wind turbines, rise significantly.

Although these rises in the prices of materials have ended up being passed on to customers and auctions for wind projects, the problem is that without benefits, companies like Siemens Gamesa find it more difficult to invest in innovation.

However, the annual growth rate of expanding the capacities of the different renewable energy sources is expected to decrease in the coming year. Even so, renewables are expected to account for more than 90% of the new generation capacity installed during that time. The largest new capacity to be installed in the next two years is expected to be located in the Asia-Pacific region. Although annual growth in China during the coming years will slow due to the exceptionally high levels of new renewable energy facilities installed in 2020, these will remain high.

In 2022, China had the largest hydropower plant capacity (413,500 MW), representing 29.7% of the world's total, generating 1,339,000 GWh or 30.4%. According to the document titled "Electricity Market Report July 2021", although electricity generation from hydropower plants in China increased by more than 3.5% in the first five months of 2021 compared to 2020, the southern and southeast of the country are facing significant droughts that significantly limit electricity generation through the use of this type of power plant. During that period, hydroelectric production decreased by more than 7% in the southern coastal provinces and about 14% in Sichuan.

However, electricity generation from hydroelectric power plants increased in Yunnan, where power demand from industries such as tin, zinc, and aluminum smelters grew significantly, but at the cost of significantly declining reservoir levels. The local grid operator asked companies to reduce power consumption in early May to avoid power cuts. In the southern province of Guangdong, a large center of industrial production, limited availability of hydropower, coupled with a strong economic recovery and associated high electricity demand, resulted in power outages and production stoppages.

Like other countries, China faces the challenges posed by the energy transition toward a low-carbon electricity system. A major problem is "gating", which means electricity generated by renewables is not adequately used by electricity grids, even when wind and solar power plants can produce it. However, most of the problem is not technical. In practice, fossil fuel power plants take precedence over renewables, leaving less room for solar and wind power in a country with vast excess coal-fired generation capacity. There is also a lack of clarity on how the renewable energy integration mandate should be enforced. Better design and implementation of appropriate policies can help. Recently, the Chinese government has launched three measures to address the problem. These measures were the following:

1. **Ban construction of new coal-fired power plants**. The central government has ordered 13 provinces to suspend approvals for new coal-fired power plants and another 15 provinces to delay the construction of new projects that have already been approved. The Chinese government has also established an early warning mechanism to anticipate and discourage local decisions that may exacerbate coal-fired power overcapacity in the future. Facing coal-fired power plants planned for the immediate future, the Chinese government has issued an alert in 28 provinces, asking local authorities to suspend their approval and companies to reconsider the investment. By curbing the development of new coal-fired power plants, the main source of fossil fuel electricity in China, the government aims to avoid competition with renewables;
2. **Implementation of management standards that guarantee the sale of renewable energy generation to the grid**. The management regulations issued by the Chinese government establish a minimum annual purchase guarantee for wind and solar generation, ensuring cost recovery and reasonable benefits for these projects. For the first time, the management rules stipulate that renewable power plants are compensated by fossil fuel power plants that squeeze out their share of guaranteed purchase, generating electricity that should have been generated from renewable sources. Above the guaranteed purchase level, renewables are encouraged to participate in the production market with competitive prices. Since renewables have low marginal costs of generation, the movement can further boost electricity generation through increased use of renewables. Other renewable energies, such as biomass, geothermal, and distributed solar energy, have guaranteed the purchase of all electricity generated;
3. **Consumption and generation objectives for renewable energies**. The new government directive sets a minimum electricity consumption from non-hydroelectric renewable energy (solar, wind, biomass, geothermal, etc.). Thirty-one provinces now have consumption targets ranging from 5% to 13% in 2020, creating incentives for local authorities to integrate renewables. At the same time, the directive obliges electricity companies to generate at least 9% of their electricity from non-hydraulic renewable sources by 2020, almost double the generation quota in the last five years. It also states the intent to develop a renewable energy certificate trading system, where power companies can sell or buy electricity generated from renewables to meet the abovementioned requirements.

The latest measures taken by the Chinese government are a significant step toward reducing the waste of electricity generated by renewables in China. They also reflect the government's attempt to harness the power of the market to meet this challenge. Going beyond the goal to be achieved by 2030, a group of energy experts has indicated that China has a fighting chance to achieve a high penetration of renewable energy in the immediate future, where renewable energy could reach more than 60% of primary energy consumption, and 85% of electricity consumption by 2050. But, these targets can only be achieved if China pursues aggressive policies to support using renewable energy for power generation for years to come. The latest actions taken by the Chinese government give reason to think that China could be on the right track to change its current energy matrix, giving greater weight to using renewable energy for electricity generation and closing down old and inefficient coal-fired power plants, thus reducing the high pollution in many of its main cities and regions.

Finally, it is important to highlight that China and the United States continue to attract around half of the investments in the global energy sector, showing positive investment growth in 2020 despite the pandemic. This growth was driven by a large increase in spending on renewable energy projects—especially wind—before the withdrawal of the previous subsidy regime in China and a deadline for eligibility for tax credits in the United States. After a record year, investments in renewable energy fell in China in 2021, even though the country announced a goal of carbon neutrality to be reached by 2060.

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